

Azure Data Factory (ADF)

Cloud Data Integration Service

Table of Contents

1. Introduction to Azure Data Factory
2. Why Use ADF
3. Core Concepts of ADF
4. ADF Architecture
5. Azure Data Factory Components
6. Azure Portal Basics
7. Creating Azure Free Account
8. Creating Azure Data Factory Instance
9. Understanding Integration Runtime
10. Linked Services
11. Datasets
12. Pipelines
13. Activities in ADF
14. Copy Data Activity
15. Data Flow
16. Control Flow Activities
17. Parameters and Variables
18. Expressions and Dynamic Content
19. Triggers
20. Monitoring and Debugging
21. Error Handling
22. Incremental Load
23. Full Load
24. ETL vs ELT in ADF
25. Connecting Different Data Sources
26. SQL Integration with ADF
27. File Handling in ADF
28. REST API Integration
29. Scheduling Pipelines
30. CI/CD in ADF

31. Git Integration
32. ARM Templates
33. Security in ADF
34. Managed Identity
35. Role-Based Access Control (RBAC)
36. Azure Key Vault Integration
37. Performance Optimization
38. Logging and Monitoring
39. Data Lake Integration
40. Synapse Integration
41. Databricks Integration
42. Snowflake Integration
43. SAP Integration
44. Real-Time Scenarios
45. End-to-End Projects
46. Best Practices
47. Common Interview Questions
48. ADF Developer Roadmap
49. ADF Naming Standards
50. Important Terminologies
51. Common Errors and Fixes
52. Certification Guidance
53. Troubleshooting and Common Issues
54. Advanced Topics and Future Trends
55. Learning Resources
56. Conclusion

1. Introduction to Azure Data Factory

Azure Data Factory (ADF) is Microsoft's fully managed, cloud-based data integration service that enables you to create, schedule, and orchestrate data pipelines for ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) workflows.

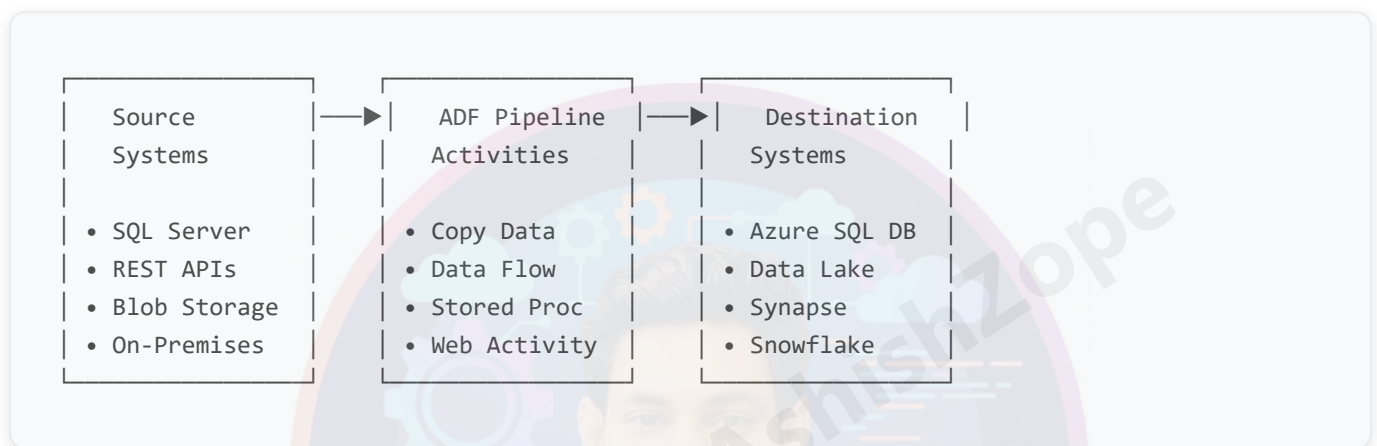
Key Capabilities

- **Data Movement:** Copy data between 100+ data stores at scale
- **Data Transformation:** Transform data using visual data flows or compute services
- **Orchestration:** Schedule and monitor complex data workflows
- **Hybrid Integration:** Connect to on-premises and cloud data sources
- **Code-Free Development:** Low-code/no-code interface with visual authoring

Real-World Use Cases

- **Data Warehousing:** Loading operational data into Azure Synapse Analytics
- **Data Lake Ingestion:** Processing streaming data into Azure Data Lake Storage
- **API Integration:** Ingesting data from REST APIs into databases
- **Cross-Cloud Analytics:** Moving data between AWS, GCP, and Azure
- **Real-Time Processing:** Event-driven data pipelines with Azure Event Grid

Example Architecture



Target Audience

- **Data Engineers:** Design and implement data pipelines
- **BI Developers:** Build data transformation workflows
- **Cloud Architects:** Design enterprise data architectures
- **Analytics Teams:** Prepare data for reporting and ML
- **DevOps Engineers:** Implement CI/CD for data pipelines

2. Why Use ADF

Advantages of ADF

Scalability & Performance

- **Elastic Compute:** Auto-scale based on workload demands
- **Parallel Processing:** Execute multiple activities simultaneously
- **Data Integration Units (DIU):** Scale compute power from 2 to 256 DIUs

- **Global Distribution:** Deploy factories in 40+ Azure regions

Cost Optimization

- **Pay-per-Use:** Only pay for actual data movement and execution time
- **Serverless:** No infrastructure management costs
- **Reserved Instances:** Up to 65% savings with 1-year reservations
- **Auto-Pause:** Automatically pause pipelines to reduce costs

Security & Compliance

- **Enterprise Security:** SOC 2, HIPAA, GDPR compliant
- **Encryption:** Data encrypted at rest and in transit
- **Managed Identity:** Passwordless authentication
- **Private Endpoints:** Secure connectivity without public IPs

Developer Experience

- **Visual Authoring:** Drag-and-drop pipeline design
- **Code Integration:** ARM templates, PowerShell, .NET SDK
- **Git Integration:** Source control with Azure DevOps/GitHub
- **CI/CD Support:** Automated deployment pipelines

Real-Time Use Cases with Examples

E-Commerce Data Pipeline

Daily Sales Data Flow:

Shopify API → ADF Pipeline → Azure SQL DB → Power BI Reports

- Extract orders via REST API
- Transform currency and dates
- Load into dimensional model
- Trigger Power BI dataset refresh

Financial Services

Banking Transaction Processing:

On-prem Oracle DB → Self-Hosted IR → Azure Synapse → Compliance Reports

- Incremental load of transactions
- Data validation and cleansing

- Load into data warehouse
- Generate regulatory reports

IoT Analytics

Sensor Data Ingestion:

IoT Hub → Event Grid → ADF Pipeline → Data Lake → Databricks

- Real-time event triggers
- Batch processing of sensor data
- Store in Parquet format
- ML model scoring pipeline

Healthcare Data Integration

Patient Records Pipeline:

EHR Systems → ADF → Azure SQL DB → Azure Analysis Services

- HIPAA-compliant data movement
- Patient data de-identification
- Load into analytics database
- Power BI embedded reports

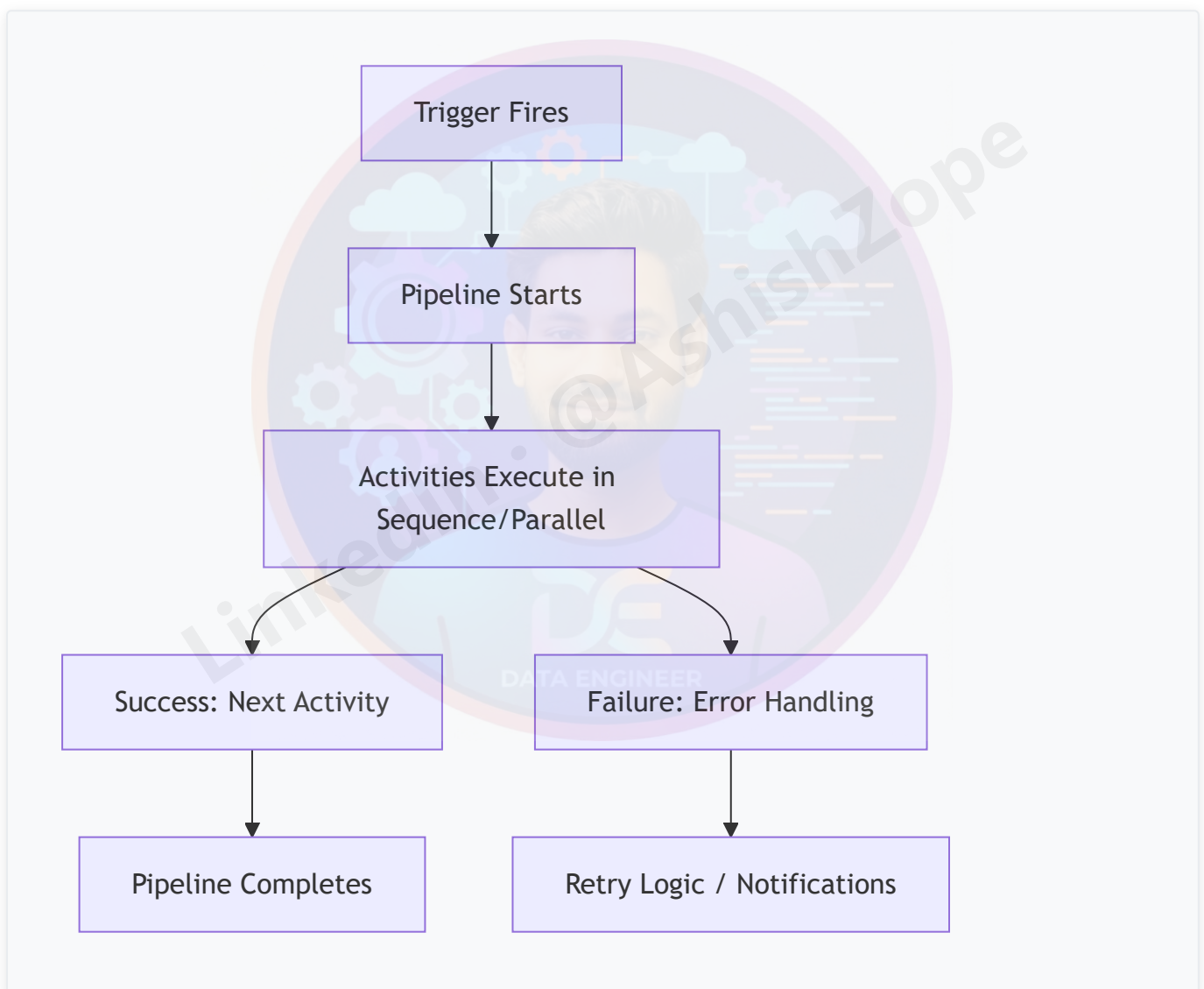
3. Core Concepts of ADF

Main Building Blocks

COMPONENT	DESCRIPTION	EXAMPLE
Pipeline	Logical grouping of activities that perform a unit of work	<code>PL_Daily_Sales_Load</code> - Orchestrates data extraction, transformation, and loading
Activity	Single processing step within a pipeline	<code>Copy_Orders_From_API</code> - Copies data from REST API to Azure SQL
Dataset	Named reference to data structure	<code>DS_Orders_JSON</code> - Points to JSON files in Blob Storage
Linked Service	Connection definition with authentication	<code>LS_SQLDW_Prod</code> - Connection string to Azure Synapse Analytics

COMPONENT	DESCRIPTION	EXAMPLE
Trigger	Mechanism to execute pipelines	<code>TR_Hourly_Data_Load</code> - Runs pipeline every hour
Integration Runtime	Compute infrastructure for data movement	<code>IR_Azure_AutoResolve</code> - Serverless compute for cloud data
Data Flow	Visual data transformation engine	<code>DF_Cleanse_Customer_Data</code> - Transforms raw customer data

Pipeline Execution Flow



Example: Complete Pipeline Structure

```
{  
  "name": "PL_Process_Sales_Data",
```

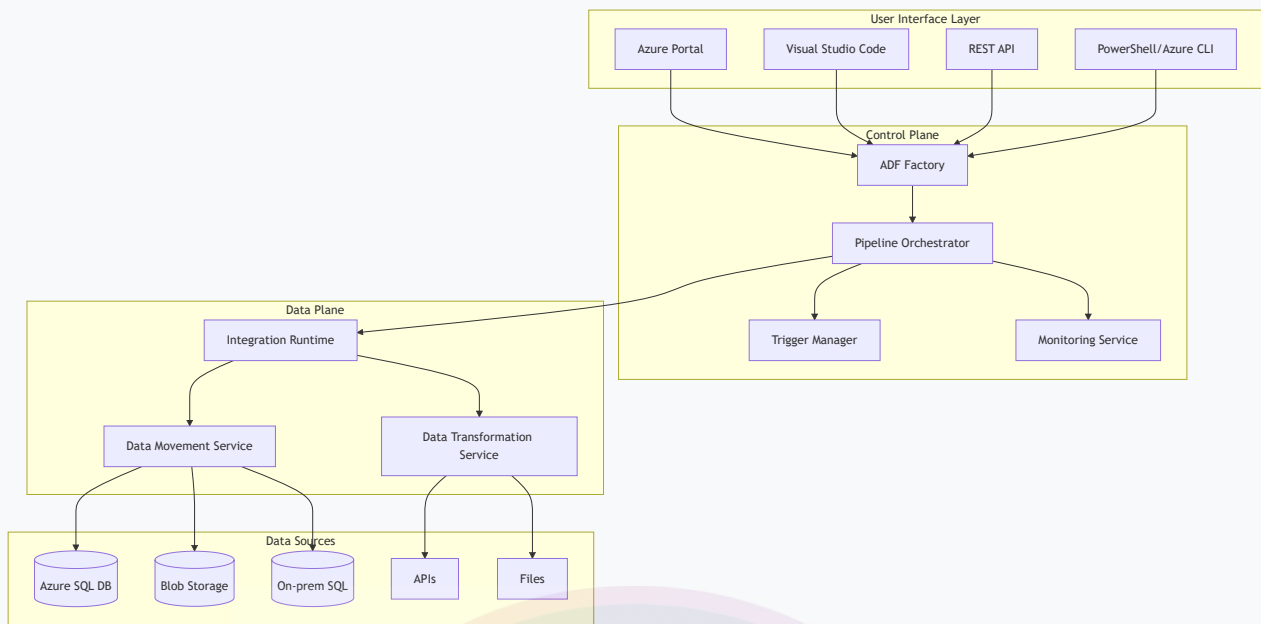
```
"properties": {
  "activities": [
    {
      "name": "Copy_Raw_Sales",
      "type": "Copy",
      "inputs": [{"referenceName": "DS_Sales_CSV"}],
      "outputs": [{"referenceName": "DS_Staging_SQL"}]
    },
    {
      "name": "Validate_Data",
      "type": "SqlServerStoredProcedure",
      "linkedServiceName": {"referenceName": "LS_SQL_Server"}
    },
    {
      "name": "Send_Notification",
      "type": "WebActivity",
      "dependsOn": [
        {"activity": "Validate_Data", "dependencyConditions": ["Succeeded"]}
      ]
    }
  ]
}
```

4. ADF Architecture

ADF Architecture Overview

Azure Data Factory follows a layered architecture that separates concerns and enables scalable data integration.

Architecture Diagram



ADF Architecture Flow

Azure Data Factory			
Source System → Linked Service → Dataset → Pipeline → Activity			
• SQL Server	• Connection	• Structure	• Workflow
• Oracle	• Auth Config	• Schema	• Activities
• SAP	• Credentials	• Metadata	• Scheduling
• REST API	• Endpoints	• Parameters	• Monitoring
• Blob Storage	• Keys	• Dynamic	• Triggers
• AWS S3	• Certificates	• Validation	• Alerts

Layers Breakdown

Source Layer (Data Ingestion)

- **Relational Databases:** SQL Server, Oracle, MySQL, PostgreSQL
- **NoSQL Databases:** Cosmos DB, MongoDB, Cassandra
- **Cloud Storage:** Azure Blob, AWS S3, Google Cloud Storage
- **File Systems:** FTP, SFTP, HDFS
- **APIs:** REST, GraphQL, SOAP

- **Big Data:** HBase, Hive, Spark
- **ERP/CRM:** SAP, Salesforce, Dynamics 365

Processing Layer (Data Transformation)

- **Copy Activity:** High-performance data movement
- **Data Flow:** Visual ETL transformations
- **Databricks:** Apache Spark processing
- **Stored Procedures:** SQL-based transformations
- **HDInsight:** Hadoop ecosystem processing
- **Azure Functions:** Serverless compute
- **Machine Learning:** Azure ML model scoring

Destination Layer (Data Storage)

- **Data Warehouses:** Azure Synapse, Snowflake, Redshift
- **Data Lakes:** Azure Data Lake Storage, S3
- **Databases:** Azure SQL DB, SQL MI, PostgreSQL
- **Analytics:** Azure Analysis Services, Power BI
- **Search:** Azure Cognitive Search
- **Archive:** Azure Archive Storage

Integration Runtime Types

Azure Integration Runtime (AutoResolve)

- **Location:** Cloud-based, globally distributed
- **Use Case:** Cloud-to-cloud data movement
- **Scaling:** Auto-scales based on workload
- **Security:** Managed by Microsoft
- **Cost:** Pay-per-use, no infrastructure management

Self-Hosted Integration Runtime (SHIR)

- **Location:** Customer-managed VMs or containers
- **Use Case:** Hybrid data integration, on-premises access
- **Connectivity:** Behind firewalls, private networks
- **Security:** Customer-controlled security
- **Cost:** VM/container costs + data movement fees

Azure-SSIS Integration Runtime

- **Location:** Azure VMs running SSIS
- **Use Case:** Lift-and-shift SSIS packages
- **Compatibility:** Full SSIS feature support
- **Scaling:** Scale-out with multiple nodes
- **Cost:** VM costs + SSIS licensing

5. Azure Data Factory Components

5.1 Pipelines

A pipeline is a logical grouping of activities that together perform a task. Pipelines can be executed independently or chained together.

Pipeline Characteristics

- **Modular Design:** Reusable across different scenarios
- **Parameterization:** Accept runtime parameters
- **Error Handling:** Built-in retry and failure handling
- **Monitoring:** Detailed execution logs and metrics
- **Scheduling:** Trigger-based or manual execution

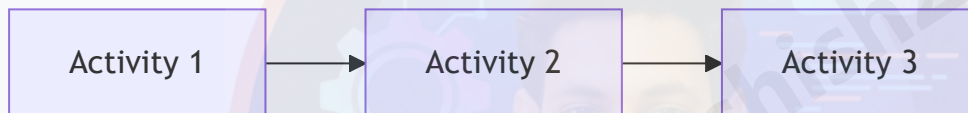
Example Pipeline Structure

```
{
  "name": "PL_Load_Customer_Data",
  "properties": {
    "description": "Daily customer data loading pipeline",
    "activities": [
      {
        "name": "Extract_Customers",
        "type": "Copy",
        "description": "Copy customer data from source to staging"
      },
      {
        "name": "Validate_Data_Quality",
        "type": "SqlServerStoredProcedure",
        "description": "Execute data validation stored procedure"
      },
      {
        "name": "Load_Dim_Customers",
        "type": "Copy",
        "description": "Load validated data to dimension table"
      }
    ]
  }
}
```

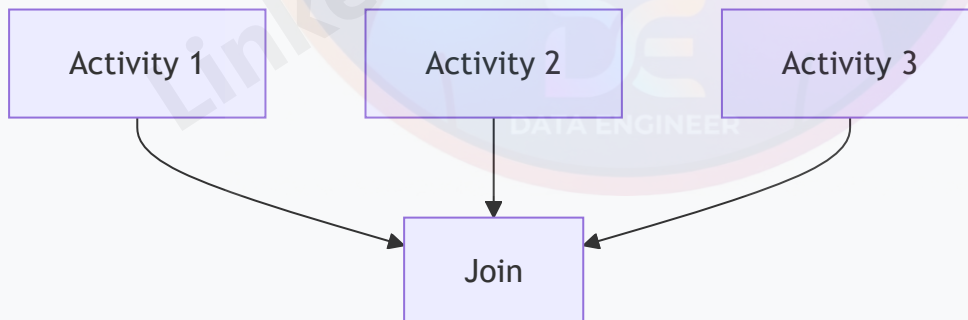
```
],  
"parameters": {  
  "SourceSystem": {  
    "type": "string",  
    "defaultValue": "CRM"  
  },  
  "LoadDate": {  
    "type": "string",  
    "defaultValue": "@utcnow()"  
  }  
}  
}  
}
```

Pipeline Execution Patterns

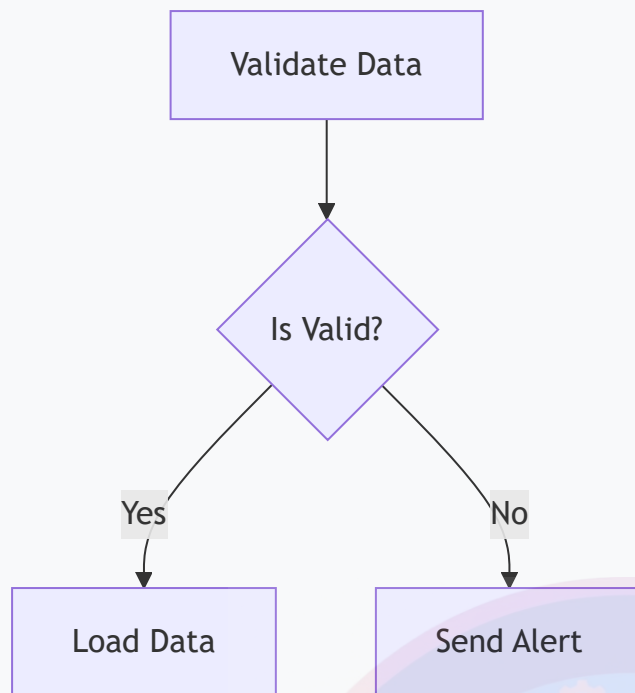
Sequential Execution



Parallel Execution



Conditional Execution



5.2 Activities

Activities are the processing steps within a pipeline. Each activity performs a specific operation on data.

Types of Activities

ACTIVITY TYPE	PURPOSE	EXAMPLE USE CASE
Copy Activity	Move data between stores	Copy CSV files to Azure SQL DB
Data Flow	Visual data transformation	Cleanse and transform customer data
Lookup	Read small reference data	Get configuration parameters
ForEach	Iterate over collections	Process multiple files in a folder
If Condition	Conditional logic	Branch based on data validation results
Execute Pipeline	Call another pipeline	Modular pipeline design
Stored Procedure	Execute SQL procedures	Data validation or audit logging
Web Activity	Call REST APIs	Send notifications or trigger external processes
Wait	Add delays	Rate limiting or scheduling dependencies
Fail	Intentionally fail pipeline	Custom error handling scenarios

Activity Dependencies

Activities can be chained with dependency conditions:

- **Succeeded:** Execute only if previous activity succeeds
- **Failed:** Execute only if previous activity fails
- **Completed:** Execute regardless of previous activity status
- **Skipped:** Execute if previous activity is skipped

Example: Complex Activity Flow

```
{
  "name": "Process_Sales_Data",
  "type": "Copy",
  "dependsOn": [
    {
      "activity": "Validate_Source_Data",
      "dependencyConditions": ["Succeeded"]
    }
  ],
  "policy": {
    "timeout": "01:00:00",
    "retry": 3,
    "retryIntervalInSeconds": 30
  }
}
```

5.3 Datasets

Datasets represent the structure of data in linked data stores. They define the schema, location, and access patterns for data.

Dataset Types

DATASET TYPE	DESCRIPTION	EXAMPLE
DelimitedText	CSV, TSV files	<code>sales_data.csv</code>
JSON	JSON files/documents	<code>api_response.json</code>
Parquet	Columnar storage format	<code>analytics_data.parquet</code>
SQL Table	Database tables	<code>dbo.Customers</code>
Binary	Binary files	<code>images.zip</code>

DATASET TYPE	DESCRIPTION	EXAMPLE
REST	REST API endpoints	<code>https://api.example.com/data</code>

Dataset Parameters

Datasets support parameterization for dynamic behavior:

```
{
  "name": "DS_Dynamic_SQL_Table",
  "properties": {
    "linkedServiceName": {
      "referenceName": "LS_SQL_Server",
      "type": "LinkedServiceReference"
    },
    "parameters": {
      "TableName": {
        "type": "string"
      },
      "SchemaName": {
        "type": "string",
        "defaultValue": "dbo"
      }
    },
    "type": "SqlServerTable",
    "typeProperties": {
      "tableName": {
        "value": "@concat(dataset().SchemaName, '.', dataset().TableName)",
        "type": "Expression"
      }
    }
  }
}
```

5.4 Linked Services

Linked services define the connection information to external data stores and compute services.

Authentication Methods

METHOD	DESCRIPTION	USE CASE
Connection String	Embedded credentials	Development environments
Key Vault	Secure credential storage	Production environments

METHOD	DESCRIPTION	USE CASE
Managed Identity	Azure AD authentication	Secure, passwordless access
Service Principal	App registration auth	Automated deployments
Basic Auth	Username/password	Legacy system integration
OAuth2	Token-based auth	API integrations

Example: Azure SQL Database Linked Service

```
{
  "name": "LS_Azure_SQL_DB",
  "properties": {
    "type": "AzureSqlDatabase",
    "typeProperties": {
      "connectionString": "Integrated Security=False;Encrypt=True;Connection
Timeout=30;Data Source=myserver.database.windows.net;Initial Catalog=mydatabase",
      "password": {
        "type": "AzureKeyVaultSecret",
        "store": {
          "referenceName": "LS_KeyVault",
          "type": "LinkServiceReference"
        },
        "secretName": "sql-password"
      },
      "userName": {
        "type": "AzureKeyVaultSecret",
        "store": {
          "referenceName": "LS_KeyVault",
          "type": "LinkServiceReference"
        },
        "secretName": "sql-username"
      }
    }
  }
}
```

5.5 Triggers

Triggers define how and when pipelines should be executed.

Trigger Types

TRIGGER TYPE	DESCRIPTION	EXAMPLE
Schedule	Time-based execution	Daily at 6 AM
Tumbling Window	Process data in time windows	Hourly batches
Event	Event-driven execution	File arrival in Blob Storage
Manual	On-demand execution	User-initiated runs

Example: Schedule Trigger

```
{
  "name": "TR_Daily_Load",
  "properties": {
    "type": "ScheduleTrigger",
    "typeProperties": {
      "recurrence": {
        "frequency": "Day",
        "interval": 1,
        "startTime": "2024-01-01T06:00:00Z",
        "timeZone": "UTC"
      }
    }
  },
  "pipelines": [
    {
      "pipelineReference": {
        "referenceName": "PL_Daily_Data_Load",
        "type": "PipelineReference"
      },
      "parameters": {
        "LoadDate": "@trigger().scheduledTime"
      }
    }
  ]
}
```

6. Azure Portal Basics

Important Azure Services

SERVICE	PURPOSE
Resource Group	Container for resources

SERVICE	PURPOSE
Storage Account	Store files/data
Azure SQL Database	Database service
Key Vault	Secrets management
Virtual Network	Networking
Monitor	Logging and monitoring

7. Creating Azure Free Account

Steps

1. Go to Azure official website
2. Create Microsoft account
3. Verify mobile number
4. Add card details
5. Activate free subscription

Free Services

- Free credits
- Limited free storage
- Free databases
- Limited compute

8. Creating Azure Data Factory Instance

Steps

1. Login to Azure Portal
2. Search for Azure Data Factory
3. Click Create
4. Select subscription
5. Select resource group
6. Enter factory name

7. Select region
8. Click Review + Create
9. Click Create

9. Understanding Integration Runtime

Integration Runtime (IR) is the compute infrastructure used by ADF.

Types of IR

TYPE	PURPOSE
Azure IR	Cloud data movement
Self-Hosted IR	On-premise connectivity
Azure SSIS IR	Execute SSIS packages

Self-Hosted IR Usage

Used when:

- Data is inside local servers
- Firewall restrictions exist
- Private network access required

10. Linked Services

Linked service acts like a connection manager.

Common Linked Services

- Azure Blob Storage
- Azure SQL Database
- SQL Server
- Oracle
- Snowflake
- REST API
- Amazon S3

- FTP/SFTP

Steps to Create Linked Service

1. Open Manage Hub
 2. Click Linked Services
 3. Click New
 4. Select connector
 5. Enter credentials
 6. Test connection
 7. Create
-

11. Datasets

Dataset defines data structure.

Dataset Examples

DATASET TYPE	EXAMPLE
Delimited Text	CSV file
JSON	API response
Parquet	Optimized analytics file
SQL Table	Employee table

Parameters in Dataset

Datasets can use dynamic values.

Example:

- Dynamic file names
 - Dynamic table names
 - Dynamic folder paths
-

12. Pipelines

Pipeline is the main orchestration component.

Pipeline Features

- Sequential execution
- Parallel execution
- Conditional execution
- Dynamic execution
- Scheduling

Real-Time Example

1. Read CSV file
 2. Validate data
 3. Load to staging table
 4. Execute stored procedure
 5. Send success email
-

13. Activities in ADF

13.1 Copy Activity

Used to move data.

Supported Sources

- SQL
- Oracle
- Blob
- S3
- FTP
- APIs

Supported Destinations

- Synapse
- SQL Database
- Data Lake

- Snowflake

13.2 Lookup Activity

Used to fetch small amounts of data.

13.3 Get Metadata Activity

Used to retrieve:

- File names
- File size
- Last modified date

13.4 Delete Activity

Deletes files/folders.

13.5 Validation Activity

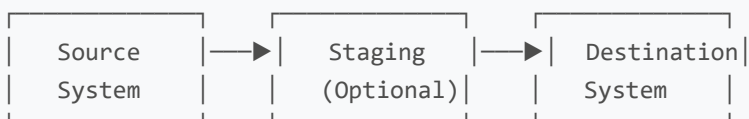
Validates file existence.

14. Copy Data Activity

Copy Activity is the most fundamental and widely used activity in ADF, designed for high-performance data movement between various data stores.

Working Flow

The copy activity follows a three-stage process:



Stage 1: Read from Source

- Connect to source data store
- Apply source filters and partitioning
- Read data in parallel using multiple threads

Stage 2: Staging (Optional)

- Temporary storage for data transformation
- Useful for cross-region data movement
- Enables data format conversion

Stage 3: Write to Destination

- Connect to destination data store
- Apply data mapping and transformations
- Write data with error handling

Supported Sources and Destinations

Source Systems

- **Databases:** SQL Server, Oracle, MySQL, PostgreSQL, DB2
- **Data Warehouses:** Azure Synapse, Snowflake, Redshift
- **Cloud Storage:** Azure Blob, AWS S3, Google Cloud Storage
- **File Systems:** FTP, SFTP, HDFS
- **APIs:** REST, GraphQL endpoints
- **Big Data:** Hive, Spark, Cosmos DB

Destination Systems

- **Databases:** Azure SQL DB, SQL MI, PostgreSQL
- **Data Warehouses:** Azure Synapse, Snowflake
- **Data Lakes:** Azure Data Lake Storage Gen2
- **Analytics:** Azure Analysis Services
- **Search:** Azure Cognitive Search

Copy Activity Settings

Source Configuration


```
{
  "source": {
    "type": "DelimitedTextSource",
    "storeSettings": {
      "type": "AzureBlobStorageReadSettings",
      "recursive": true,
      "wildcardFolderPath": "data/year=*/month=*",
      "wildcardFileName": "*.csv"
    },
    "formatSettings": {
      "type": "DelimitedTextReadSettings",
      "skipLineCount": 1
    }
  }
}
```

Sink Configuration

```
{
  "sink": {
    "type": "SqlSink",
    "writeBehavior": "insert",
    "sqlWriterUseTableLock": false,
    "tableOption": "autoCreate",
    "storeSettings": {
      "type": "SqlServerStoreWriteSettings"
    }
  }
}
```

Mapping Configuration

```
{
  "mappings": [
    {
      "source": {
        "name": "customer_id",
        "type": "String"
      },
      "sink": {
        "name": "CustomerID",
        "type": "Int32"
      }
    },
    {
```

```
"source": {
  "name": "full_name",
  "type": "String"
},
"sink": {
  "name": "CustomerName",
  "type": "String"
}
}
]
```

Performance Optimization

Parallel Copy Settings

- **Degree of Copy Parallelism:** Number of parallel threads (default: 1)
- **Data Integration Units (DIU):** Compute power allocation (2-256)
- **Parallel Copies:** Maximum parallel copies per activity

Partitioning Options

- **Physical Partitioning:** Split data by physical partitions
- **Dynamic Range Partitioning:** Split by column values
- **Hash Partitioning:** Distribute data evenly across partitions

Example: Optimized Copy Activity

```
{
  "name": "Copy_Large_Dataset",
  "type": "Copy",
  "typeProperties": {
    "source": {
      "type": "ParquetSource",
      "storeSettings": {
        "type": "AzureBlobStorageReadSettings",
        "enablePartitionDiscovery": true
      }
    },
    "sink": {
      "type": "SqlSink",
      "writeBehavior": "upsert",
      "upsertSettings": {
        "useTempDB": true,
        "keys": ["CustomerID"]
      }
    }
  }
}
```

```
}  
,  
"translator": {  
  "type": "TabularTranslator",  
  "mappings": [  
    {  
      "source": {"name": "customer_id"},  
      "sink": {"name": "CustomerID"}  
    }  
  ]  
}  
,  
"policy": {  
  "timeout": "02:00:00",  
  "retry": 3,  
  "retryIntervalInSeconds": 30  
}  
}
```

Fault Tolerance Features

Error Handling

- **Fault Tolerance:** Skip incompatible rows
- **Redirect Incompatible Rows:** Store bad data separately
- **Logging Level:** Control error logging verbosity

Retry Logic

- **Retry Count:** Number of retry attempts
- **Retry Interval:** Time between retries
- **Retryable Errors:** Network timeouts, temporary failures

Copy Activity Best Practices

Performance Tips

1. **Use Staging:** For cross-region transfers
2. **Enable Compression:** Reduce network bandwidth
3. **Optimize DIUs:** Scale compute based on data volume
4. **Use Partitioning:** Parallelize large dataset copies
5. **Monitor Throughput:** Track performance metrics

Data Quality Tips

1. **Validate Schemas:** Ensure source/destination compatibility
2. **Handle Data Types:** Proper type mapping and conversion
3. **Manage Null Values:** Define null handling strategies
4. **Set Row Limits:** Prevent runaway processing

Example: Complete Copy Pipeline

```
{
  "name": "PL_Copy_With_Error_Handling",
  "properties": {
    "activities": [
      {
        "name": "Copy_Sales_Data",
        "type": "Copy",
        "inputs": [
          { "referenceName": "DS_Sales_CSV", "type": "DatasetReference" }
        ],
        "outputs": [
          { "referenceName": "DS_Sales_SQL", "type": "DatasetReference" }
        ],
        "typeProperties": {
          "source": {
            "type": "DelimitedTextSource",
            "storeSettings": {
              "type": "AzureBlobStorageReadSettings"
            },
            "formatSettings": {
              "type": "DelimitedTextReadSettings",
              "skipLineCount": 1
            }
          },
          "sink": {
            "type": "SqlSink",
            "writeBehavior": "insert",
            "sqlWriterStoredProcedureName": "usp_InsertSalesData",
            "sqlWriterTableType": "SalesDataType",
            "storeSettings": {
              "type": "SqlServerStoreWriteSettings"
            }
          },
          "enableStaging": false,
          "translator": {
            "type": "TabularTranslator",
            "mappings": [
              {
                "source": { "name": "OrderID", "type": "String" },
                "sink": { "name": "OrderID", "type": "Int32" }
              }
            ]
          }
        }
      }
    ]
  }
}
```

```
{
  "source": {"name": "OrderDate", "type": "String"},
  "sink": {"name": "OrderDate", "type": "DateTime"}
}
],
},
"policy": {
  "timeout": "01:00:00",
  "retry": 3,
  "retryIntervalInSeconds": 30
}
}
]
```

15. Data Flow

Data Flow is ADF's visual data transformation engine that enables code-free ETL/ELT operations using a drag-and-drop interface.

Types of Data Flows

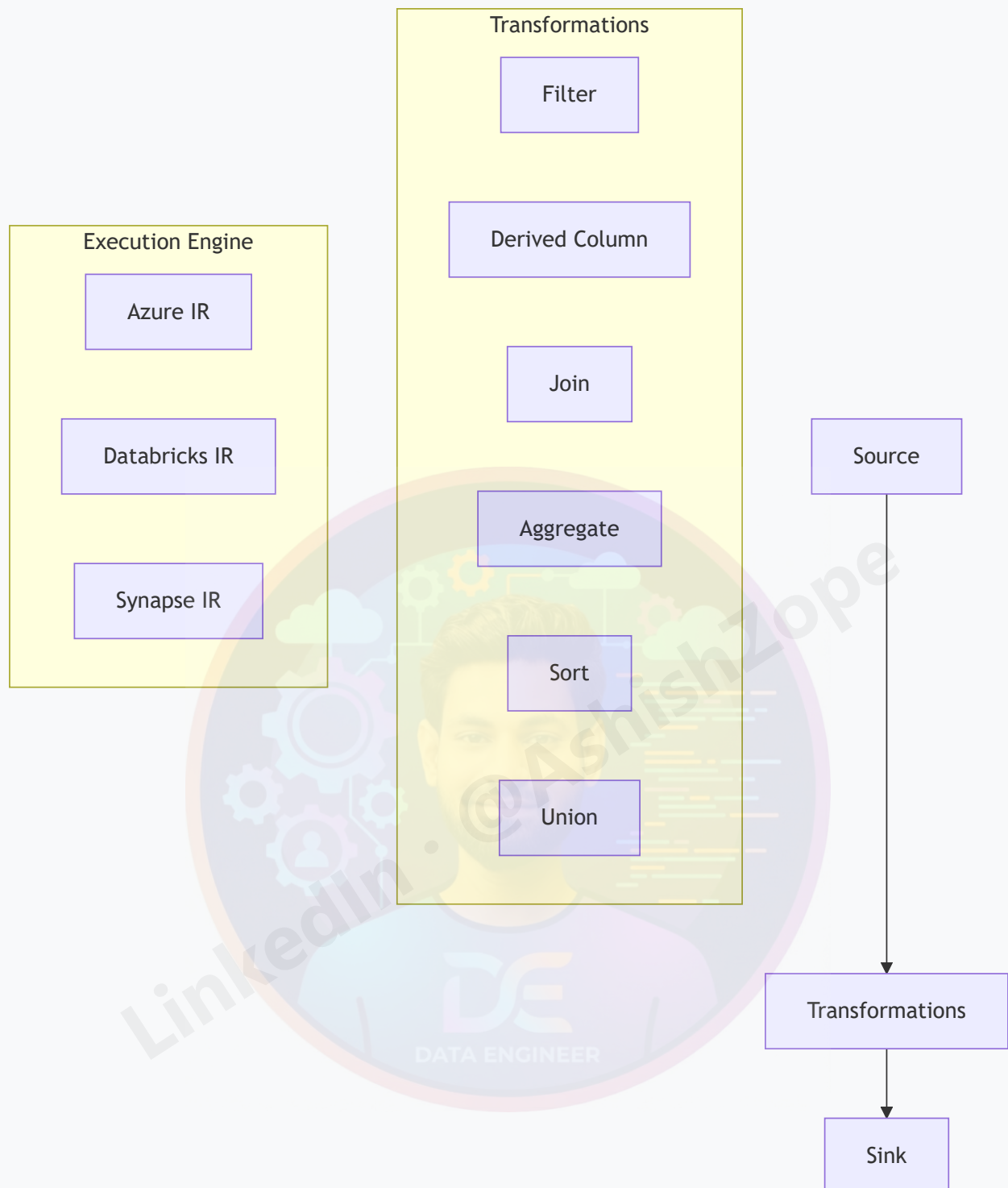
Mapping Data Flow

- **Purpose:** Visual ETL transformations
- **Execution:** Spark-based processing
- **Use Case:** Complex data transformations
- **Target:** Data warehouses, data lakes

Wrangling Data Flow

- **Purpose:** Power Query-style transformations
- **Execution:** Spark-based processing
- **Use Case:** Data preparation and cleansing
- **Target:** Self-service data prep

Data Flow Architecture



Common Transformations

Data Cleansing Transformations

TRANSFORMATION	PURPOSE	EXAMPLE
Filter	Remove unwanted rows	<code>WHERE status != 'inactive'</code>
Derived Column	Create calculated columns	<code>FullName = concat(FirstName, ' ', LastName)</code>
Select	Choose specific columns	Keep only relevant fields
Sort	Order data	Sort by date descending
Distinct	Remove duplicates	Unique customer records

Data Integration Transformations

TRANSFORMATION	PURPOSE	EXAMPLE
Join	Combine datasets	Customer + Orders = Sales data
Union	Stack datasets	Combine monthly sales files
Lookup	Enrich with reference data	Add product categories
Exists	Check for matches	Validate against master data
Conditional Split	Route data conditionally	Valid vs invalid records

Data Aggregation Transformations

TRANSFORMATION	PURPOSE	EXAMPLE
Aggregate	Group and summarize	Sales by region and month
Window	Rolling calculations	Moving averages, rankings
Pivot	Crosstab data	Months as columns
Unpivot	Normalize data	Columns to rows

Data Flow Execution Modes

Debug Mode

- **Purpose:** Test and validate transformations
- **Data Sampling:** Process subset of data
- **Interactive:** Real-time preview of results
- **Cost:** Minimal compute usage

Pipeline Execution

- **Purpose:** Production data processing
- **Full Dataset:** Process all data
- **Performance:** Optimized for large volumes
- **Integration:** Part of pipeline workflows

Example: Complete Data Flow

```
{
  "name": "DF_Transform_Sales_Data",
  "properties": {
    "type": "MappingDataFlow",
    "typeProperties": {
      "sources": [
        {
          "name": "SalesSource",
          "dataset": {
            "referenceName": "DS_Sales_CSV",
            "type": "DatasetReference"
          }
        }
      ],
      "sinks": [
        {
          "name": "SalesSink",
          "dataset": {
            "referenceName": "DS_Sales_Parquet",
            "type": "DatasetReference"
          }
        }
      ],
      "transformations": [
        {
          "name": "FilterValidRecords",
          "type": "Filter",
          "typeProperties": {
            "condition": "status == 'active' && amount > 0"
          }
        },
        {
          "name": "AddCalculatedColumns",
          "type": "DerivedColumn",
          "typeProperties": {
            "columns": [
              {
                "name": "TotalAmount",
                "expression": "amount * quantity"
              }
            ]
          }
        }
      ]
    }
  }
}
```

```
{
  "name": "OrderDateFormatted",
  "expression": "toDate(order_date, 'yyyy-MM-dd')"
}
],
{
  "name": "JoinWithCustomers",
  "type": "Join",
  "typeProperties": {
    "left": "FilterValidRecords",
    "right": "CustomerLookup",
    "joinType": "inner",
    "leftKeys": ["customer_id"],
    "rightKeys": ["id"]
  }
},
],
"script": "source(output(..."
```

Data Flow Performance Optimization

Cluster Configuration

- **Compute Type:** General Purpose vs Memory Optimized
- **Core Count:** Number of Spark cores
- **Time to Live:** Cluster idle timeout

Partitioning Strategy

- **Current Partitioning:** Maintain existing partitions
- **Hash Partitioning:** Distribute evenly across partitions
- **Key Partitioning:** Partition by specific columns
- **Round Robin:** Random distribution

Caching Strategy

- **Output Caching:** Cache transformation results
- **Input Caching:** Cache source data
- **Memory Management:** Optimize for large datasets

Data Flow Best Practices

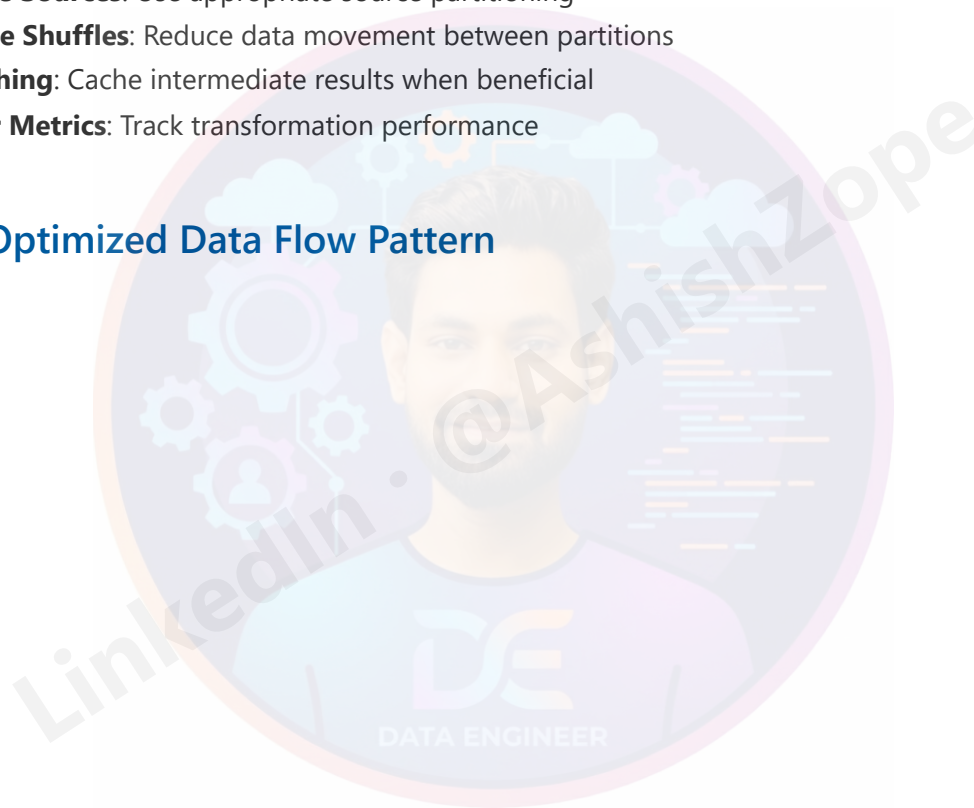
Development Best Practices

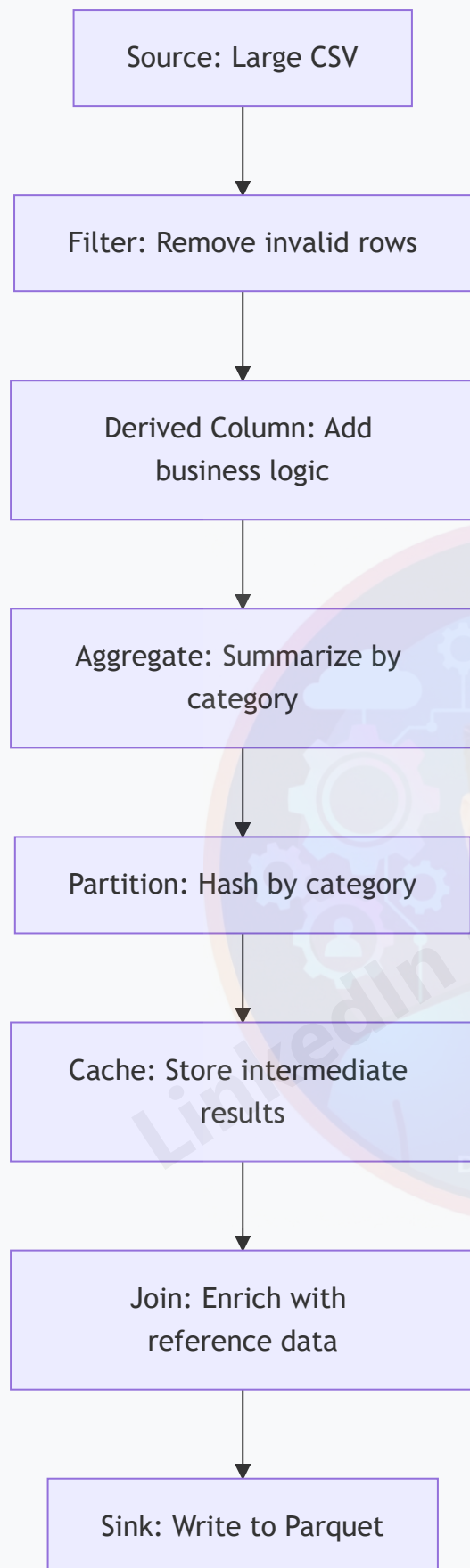
1. **Modular Design:** Break complex flows into smaller flows
2. **Naming Conventions:** Use descriptive transformation names
3. **Data Preview:** Regularly preview data at each step
4. **Error Handling:** Implement proper error handling patterns

Performance Best Practices

1. **Optimize Sources:** Use appropriate source partitioning
2. **Minimize Shuffles:** Reduce data movement between partitions
3. **Use Caching:** Cache intermediate results when beneficial
4. **Monitor Metrics:** Track transformation performance

Example: Optimized Data Flow Pattern





Integration with Other Services

Synapse Integration

- **Serverless SQL Pools:** Query data flow outputs
- **Dedicated SQL Pools:** Load transformed data
- **Spark Pools:** Execute complex transformations

Databricks Integration

- **Delta Lake:** ACID transactions on data lakes
- **MLflow:** Model training and deployment
- **Structured Streaming:** Real-time data processing

Power BI Integration

- **Direct Query:** Query data flow outputs
 - **Import Mode:** Scheduled data refreshes
 - **Composite Models:** Combine multiple data sources
-

16. Control Flow Activities

Important Control Activities

ACTIVITY	PURPOSE
If Condition	Conditional logic
Switch	Multiple conditions
Until	Loop until condition met
ForEach	Iterate collections
Wait	Delay execution
Fail	Fail pipeline intentionally

17. Parameters and Variables

Parameters and variables enable dynamic behavior in ADF pipelines, allowing for reusable and configurable data workflows.

Parameters

Parameters are external values passed to pipelines at runtime. They enable pipeline reusability across different environments and scenarios.

Parameter Types

TYPE	DESCRIPTION	EXAMPLE
String	Text values	"prod" , "@pipeline().Pipeline"
Int	Integer numbers	100 , 0
Float	Decimal numbers	1.5 , 99.99
Bool	True/false values	true , false
Array	List of values	["item1", "item2"]
Object	Key-value pairs	{"key": "value"}

Parameter Usage Examples

Pipeline Parameters

```
{
  "name": "PL_Load_Data",
  "properties": {
    "parameters": {
      "SourceContainer": {
        "type": "string",
        "defaultValue": "raw-data"
      },
      "TargetTable": {
        "type": "string",
        "defaultValue": "staging.Sales"
      },
      "LoadDate": {
        "type": "string",
        "defaultValue": "@utcnow()"
      },
      "IsIncremental": {
        "type": "bool",
        "defaultValue": false
      }
    }
  }
}
```

```

    }
  }
}

```

Dataset Parameters

```

{
  "name": "DS_Dynamic_SQL_Table",
  "properties": {
    "parameters": {
      "SchemaName": {
        "type": "string",
        "defaultValue": "dbo"
      },
      "TableName": {
        "type": "string"
      }
    },
    "type": "SqlServerTable",
    "typeProperties": {
      "tableName": {
        "value": "@concat(dataset().SchemaName, '.', dataset().TableName)",
        "type": "Expression"
      }
    }
  }
}

```

Linked Service Parameters

```

{
  "name": "LS_Parameterised_SQL",
  "properties": {
    "parameters": {
      "ServerName": {
        "type": "string"
      },
      "DatabaseName": {
        "type": "string"
      }
    },
    "type": "SqlServer",
    "typeProperties": {
      "connectionString": "Integrated Security=False;Encrypt=True;Connection
Timeout=30;Data Source=@{linkedService().ServerName};Initial
Catalog=@{linkedService().DatabaseName}"
    }
  }
}

```



```
}  
}
```

Variables

Variables are internal pipeline values that can be set and modified during pipeline execution. They are useful for storing intermediate results or configuration values.

Variable Types

TYPE	DESCRIPTION	EXAMPLE
String	Text values	"processing"
Boolean	True/false	true
Array	List of items	["file1.csv", "file2.csv"]
Object	Complex objects	{"status": "success"}

Variable Scope

- **Pipeline Scope:** Available throughout the pipeline
- **Activity Scope:** Limited to specific activities (Set Variable activity)

Variable Activities

Set Variable Activity

```
{  
  "name": "Set_Processing_Status",  
  "type": "SetVariable",  
  "typeProperties": {  
    "variableName": "ProcessingStatus",  
    "value": "InProgress"  
  }  
}
```

Append Variable Activity

```
{
  "name": "Append_File_List",
  "type": "AppendVariable",
  "typeProperties": {
    "variableName": "ProcessedFiles",
    "value": "@item().name"
  }
}
```

Expressions and Dynamic Content

ADF uses expressions to create dynamic values using functions, operators, and references.

Expression Syntax

```
@{<expression>}
```

Common Expression Functions

String Functions

FUNCTION	DESCRIPTION	EXAMPLE
<code>concat()</code>	Join strings	<code>concat('Table_', pipeline().parameters.TableSuffix)</code>
<code>substring()</code>	Extract substring	<code>substring('HelloWorld', 0, 5) → "Hello"</code>
<code>replace()</code>	Replace text	<code>replace('Hello World', 'World', 'ADF')</code>
<code>split()</code>	Split string	<code>split('a,b,c', ',') → ['a','b','c']</code>
<code>toLowerCase()</code>	Convert to lowercase	<code>toLowerCase('HELLO') → "hello"</code>
<code>toUpperCase()</code>	Convert to uppercase	<code>toUpperCase('hello') → "HELLO"</code>

Date/Time Functions

FUNCTION	DESCRIPTION	EXAMPLE
<code>utcnow()</code>	Current UTC time	<code>2024-01-15T10:30:00Z</code>
<code>adddays()</code>	Add days to date	<code>adddays(utcnow(), -1)</code>
<code>formatDateTime()</code>	Format date	<code>formatDateTime(utcnow(), 'yyyy-MM-dd')</code>
<code>startOfDay()</code>	Start of day	<code>startOfDay(utcnow())</code>
<code>dayOfWeek()</code>	Day of week	<code>dayOfWeek(utcnow())</code>

Mathematical Functions

FUNCTION	DESCRIPTION	EXAMPLE
<code>add()</code>	Addition	<code>add(10, 5) → 15</code>
<code>sub()</code>	Subtraction	<code>sub(10, 5) → 5</code>
<code>mul()</code>	Multiplication	<code>mul(10, 5) → 50</code>
<code>div()</code>	Division	<code>div(10, 5) → 2</code>
<code>mod()</code>	Modulo	<code>mod(10, 3) → 1</code>

Collection Functions

FUNCTION	DESCRIPTION	EXAMPLE
<code>length()</code>	Array length	<code>length(['a', 'b', 'c']) → 3</code>
<code>contains()</code>	Check membership	<code>contains(['a', 'b', 'c'], 'b') → true</code>
<code>union()</code>	Combine arrays	<code>union(['a', 'b'], ['c', 'd']) → ['a', 'b', 'c', 'd']</code>
<code>intersection()</code>	Common elements	<code>intersection(['a', 'b'], ['b', 'c']) → ['b']</code>
<code>first()</code>	First element	<code>first(['a', 'b', 'c']) → "a"</code>
<code>last()</code>	Last element	<code>last(['a', 'b', 'c']) → "c"</code>

Logical Functions

FUNCTION	DESCRIPTION	EXAMPLE
<code>if()</code>	Conditional logic	<code>if(equals(1, 1), 'Equal', 'Not Equal')</code>
<code>equals()</code>	Equality check	<code>equals('a', 'a') → true</code>
<code>and()</code>	Logical AND	<code>and(true, false) → false</code>

FUNCTION	DESCRIPTION	EXAMPLE
<code>or()</code>	Logical OR	<code>or(true, false) → true</code>
<code>not()</code>	Logical NOT	<code>not(true) → false</code>

Context References

REFERENCE	DESCRIPTION	EXAMPLE
<code>pipeline()</code>	Pipeline metadata	<code>pipeline().Pipeline</code>
<code>pipeline().parameters</code>	Pipeline parameters	<code>pipeline().parameters.SourceTable</code>
<code>pipeline().globalParameters</code>	Global parameters	<code>pipeline().globalParameters.Environment</code>
<code>variables()</code>	Pipeline variables	<code>variables('Counter')</code>
<code>activity()</code>	Activity output	<code>activity('LookupConfig').output</code>
<code>item()</code>	ForEach item	<code>item().name</code>
<code>trigger()</code>	Trigger metadata	<code>trigger().scheduledTime</code>

Advanced Expression Examples

Dynamic File Paths

```
{
  "folderPath": "@concat('data/', formatDateTime(utcnow(), 'yyyy'), '/',
formatDateTime(utcnow(), 'MM'), '/')"
}
```

Conditional Logic

```
{
  "tableName": "@if(equals(pipeline().parameters.Environment, 'prod'), 'dbo.Prod_Sales',
'dbo.Dev_Sales')"
```

Array Processing

```
{  
  "fileList": "@split(activity('Get_File_List').output.value, ',,')"  
}
```

Date Calculations

```
{  
  "startDate": "@formatDateTime(adddays(utcnow(), -30), 'yyyy-MM-dd')",  
  "endDate": "@formatDateTime(utcnow(), 'yyyy-MM-dd')"  
}
```

Best Practices for Parameters and Variables

Parameter Best Practices

1. **Use Descriptive Names:** `SourceContainer` vs `param1`
2. **Provide Default Values:** Enable testing without all parameters
3. **Validate Input:** Use appropriate data types
4. **Document Usage:** Comment complex expressions

Variable Best Practices

1. **Initialize Early:** Set variables at pipeline start
2. **Use Appropriate Scope:** Pipeline vs activity level
3. **Avoid Overuse:** Don't replace parameters with variables
4. **Monitor Values:** Log variable values for debugging

Expression Best Practices

1. **Test Expressions:** Use debug mode to validate
2. **Handle Nulls:** Use `coalesce()` for null handling
3. **Performance:** Cache expensive expressions in variables
4. **Readability:** Break complex expressions into steps

18. Expressions and Dynamic Content

ADF uses dynamic expressions.

Common Functions

FUNCTION	PURPOSE
concat()	Combine strings
utcNow()	Current UTC time
pipeline()	Pipeline metadata
variables()	Access variable
activity()	Activity output
substring()	Extract text

Example

```
@concat('File_',utcNow(),'.csv')
```

19. Triggers

Triggers automate execution.

19.1 Schedule Trigger

Runs pipeline at fixed intervals.

Example:

- Daily at 9 PM
- Every hour

19.2 Tumbling Window Trigger

Processes data in chunks/windows.

Used in incremental loading.

19.3 Event Trigger

Executes when:

- File arrives
 - Blob created
 - Blob deleted
-

20. Monitoring and Debugging

Monitoring Options

- Pipeline runs
- Activity runs
- Trigger runs
- Error logs

Debug Mode

Used for testing pipelines.

Monitoring Metrics

- Duration
 - Throughput
 - Queue time
 - Rows copied
-

21. Error Handling

Common Techniques

Try-Catch Pattern

Use:

- Execute pipeline
- Failure path
- Success path

Logging Errors

Store errors into:

- SQL table
- Blob storage
- Log analytics

Retry Mechanism

ADF supports retries.

Example:

- Retry count = 3
- Retry interval = 30 sec

22. Incremental Load

Incremental load copies only new or changed data.

Advantages

- Faster execution
- Lower cost
- Reduced load on systems

Common Approaches

METHOD	DESCRIPTION
Watermark	Based on date/time
CDC	Change Data Capture
Timestamp	Last modified value

METHOD	DESCRIPTION
Identity Column	Max ID tracking

Incremental Load Flow

1. Get max timestamp
2. Fetch new data
3. Load into target
4. Update watermark table

23. Full Load

Full load copies complete data every time.

Advantages

- Simpler logic
- Easy implementation

Disadvantages

- Higher execution time
- Higher cost
- Duplicate risk

24. ETL vs ELT in ADF

FEATURE	ETL	ELT
Transformation	Before loading	After loading
Performance	Slower	Faster
Compute	External	Target system
Example	SSIS	Synapse

ETL Flow

Extract → Transform → Load

ELT Flow

Extract → Load → Transform

25. Connecting Different Data Sources

Supported Sources

- SQL Server
 - Oracle
 - MySQL
 - PostgreSQL
 - SAP
 - Blob Storage
 - S3
 - Google Cloud Storage
 - REST APIs
 - Salesforce
 - ServiceNow
-

26. SQL Integration with ADF

Common SQL Operations

- Execute stored procedures
- Read SQL tables
- Insert/update data
- Merge operations

Stored Procedure Activity

Used to execute SQL procedures.

Example Use Cases

- Audit logging
 - Data validation
 - Merge operations
 - Data cleansing
-

27. File Handling in ADF

Supported File Formats

FORMAT	USAGE
CSV	Flat file
JSON	Semi-structured
XML	Hierarchical
Parquet	Analytics
Avro	Big data
Excel	Reporting

Compression Support

- ZIP
 - GZIP
 - BZIP2
-

28. REST API Integration

ADF can connect to APIs.

Steps

1. Create REST linked service
2. Configure authentication
3. Create dataset
4. Use copy activity

Authentication Types

- Anonymous
 - Basic
 - OAuth2
 - Managed Identity
-

29. Scheduling Pipelines

Scheduling Methods

METHOD	EXAMPLE
Hourly	Every 1 hour
Daily	Daily at 10 PM
Weekly	Every Sunday
Monthly	1st day of month

30. CI/CD in ADF

CI/CD means Continuous Integration and Continuous Deployment.

Benefits

- Automated deployment
- Version control
- Faster release cycles
- Reduced manual effort

Tools Used

- Azure DevOps
 - GitHub
 - ARM Templates
-

31. Git Integration

ADF supports Git integration.

Supported Repositories

- Azure DevOps Git
- GitHub

Benefits

- Version control
 - Collaboration
 - Rollback support
 - Branching
-

32. ARM Templates

ARM templates are used for deployment automation.

Uses

- Infrastructure deployment
 - Environment migration
 - Backup configuration
-

33. Security in ADF

Security Features

- Encryption
 - RBAC
 - Managed Identity
 - Private endpoints
 - Key Vault integration
-

34. Managed Identity

Managed identity removes need for storing passwords.

Benefits

- Secure authentication
 - Passwordless access
 - Easy integration
-

35. Role-Based Access Control (RBAC)

RBAC controls user permissions.

Common Roles

ROLE	ACCESS
Owner	Full access
Contributor	Modify resources
Reader	Read-only
Data Factory Contributor	ADF access

36. Azure Key Vault Integration

Key Vault stores secrets securely.

Stored Items

- Passwords
- API keys
- Connection strings
- Certificates

Advantages

- Improved security
 - Centralized secrets management
-

37. Performance Optimization

Optimization Techniques

- Parallel copy
- Partitioning
- Compression
- Incremental loading
- Proper IR selection
- Optimize data flow clusters

Performance Metrics

- DIU usage
 - Throughput
 - Queue duration
-

38. Logging and Monitoring

Monitoring Tools

- Azure Monitor
- Log Analytics

- Diagnostic logs
- Alerts

Alert Examples

- Pipeline failure
 - High execution time
 - Integration runtime offline
-

39. Data Lake Integration

ADF integrates with Azure Data Lake.

Advantages

- Scalable storage
- Cheap storage
- Big data processing

Common Architecture

Source → ADF → Data Lake → Synapse/Databricks

40. Synapse Integration

ADF works closely with Azure Synapse.

Use Cases

- Data warehouse loading
 - ELT processing
 - Analytics pipelines
-

41. Databricks Integration

ADF can execute Databricks notebooks.

Common Use Cases

- PySpark processing
 - Machine learning
 - Big data transformation
-

42. Snowflake Integration

ADF supports Snowflake connectivity.

Operations

- Data loading
 - Data extraction
 - Stored procedure execution
-

43. SAP Integration

ADF supports SAP connectors.

SAP Connectors

- SAP Table
 - SAP ECC
 - SAP HANA
 - SAP BW
-

44. Real-Time Scenarios

Banking Scenario

- Extract transactions

- Validate records
- Load to warehouse
- Generate reports

E-Commerce Scenario

- API ingestion
- Customer analytics
- Sales reporting

Healthcare Scenario

- Patient records integration
 - Daily ETL processing
-

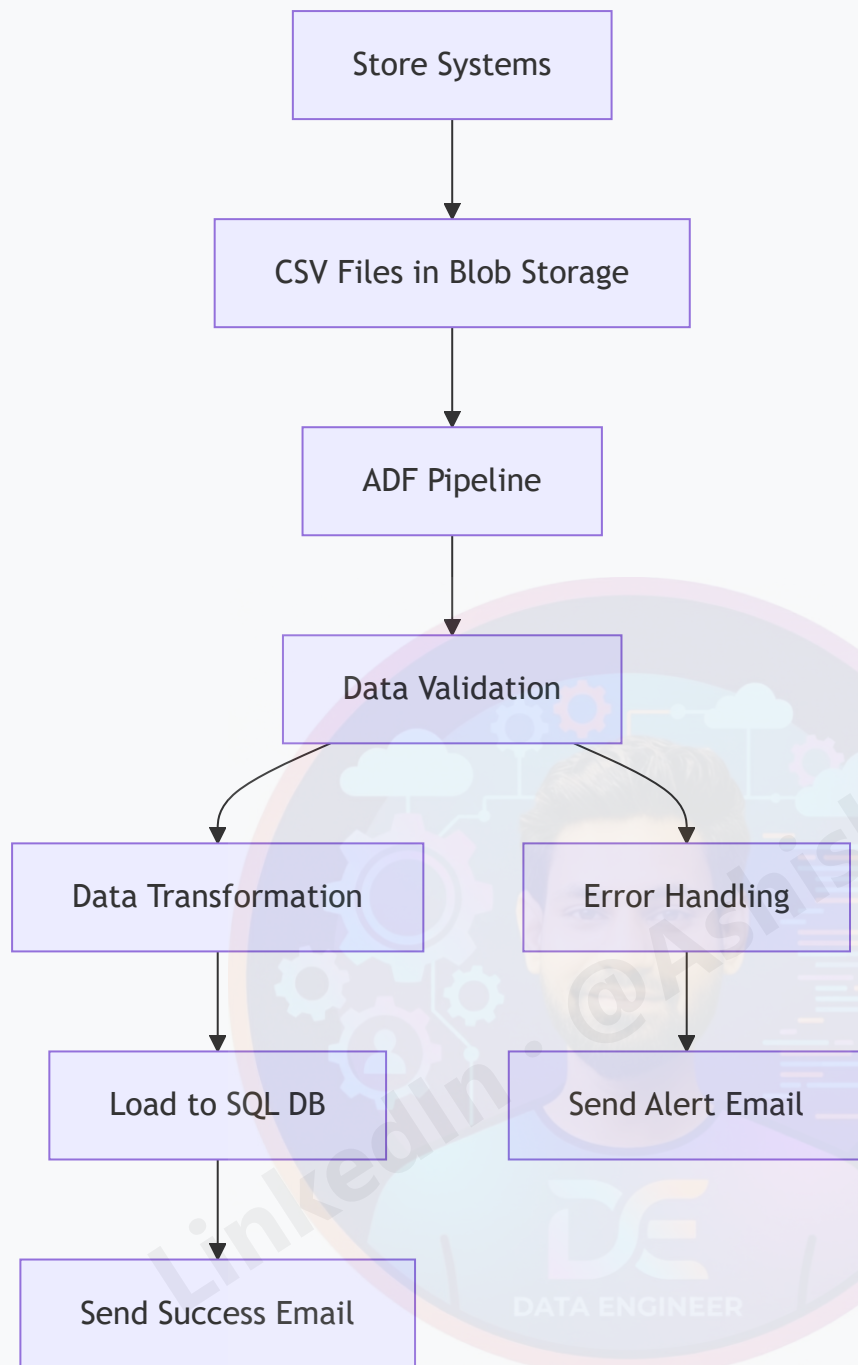
45. End-to-End Projects

Project 1: CSV to SQL Database Pipeline

Business Scenario

A retail company receives daily sales data in CSV format from multiple stores. The data needs to be validated, transformed, and loaded into a central SQL database for reporting.

Architecture Diagram



Pipeline Components

1. Linked Services

```
{
  "LS_Blob_Storage": {
    "type": "AzureBlobStorage",
    "typeProperties": {
      "connectionString": "@linkedService().StorageConnectionString"
    }
  }
}
```

```

    },
    "LS_SQL_Database": {
      "type": "AzureSqlDatabase",
      "typeProperties": {
        "connectionString": "@linkedService().SqlConnectionString"
      }
    }
  }
}

```

2. Datasets

```

{
  "DS_Sales_CSV": {
    "type": "DelimitedText",
    "linkedServiceName": {"referenceName": "LS_Blob_Storage"},
    "typeProperties": {
      "location": {
        "type": "AzureBlobStorageLocation",
        "fileName": "*.csv",
        "folderPath": "sales-data"
      },
      "columnDelimiter": ",",
      "firstRowAsHeader": true
    }
  },
  "DS_Sales_SQL": {
    "type": "AzureSqlTable",
    "linkedServiceName": {"referenceName": "LS_SQL_Database"},
    "typeProperties": {
      "tableName": "dbo.Sales"
    }
  }
}

```

3. Pipeline Activities

```

{
  "name": "PL_Load_Sales_Data",
  "properties": {
    "activities": [
      {
        "name": "Validate_File_Existence",
        "type": "GetMetadata",
        "typeProperties": {
          "dataset": {"referenceName": "DS_Sales_CSV"},
          "fieldList": ["exists", "itemName", "lastModified"]
        }
      }
    ]
  }
}

```

```

{
  "name": "Copy_Sales_Data",
  "type": "Copy",
  "dependsOn": [
    {
      "activity": "Validate_File_Existence",
      "dependencyConditions": ["Succeeded"]
    }
  ],
  "typeProperties": {
    "source": {
      "type": "DelimitedTextSource",
      "storeSettings": {"type": "AzureBlobStorageReadSettings"}
    },
    "sink": {
      "type": "SqlSink",
      "writeBehavior": "insert",
      "tableOption": "autoCreate"
    },
    "translator": {
      "type": "TabularTranslator",
      "mappings": [
        {"source": {"name": "store_id"}, "sink": {"name": "StoreID"}},
        {"source": {"name": "product_id"}, "sink": {"name": "ProductID"}},
        {"source": {"name": "sales_amount"}, "sink": {"name": "SalesAmount"}},
        {"source": {"name": "sales_date"}, "sink": {"name": "SalesDate"}}
      ]
    }
  }
},
{
  "name": "Execute_Data_Quality_Check",
  "type": "SqlServerStoredProcudure",
  "dependsOn": [
    {
      "activity": "Copy_Sales_Data",
      "dependencyConditions": ["Succeeded"]
    }
  ],
  "typeProperties": {
    "storedProcedureName": "usp_ValidateSalesData"
  }
},
{
  "name": "Send_Success_Notification",
  "type": "WebActivity",
  "dependsOn": [
    {
      "activity": "Execute_Data_Quality_Check",
      "dependencyConditions": ["Succeeded"]
    }
  ],
  "typeProperties": {
    "method": "POST",

```

```
"url": "https://api.sendgrid.com/v3/mail/send",
"headers": {
    "Authorization": "@linkedService().SendGridApiKey",
    "Content-Type": "application/json"
},
"body": {
    "personalizations": [
        {
            "to": [{"email": "data-team@company.com"}],
            "subject": "Sales Data Load Completed Successfully"
        }
    ],
    "from": {"email": "adf@company.com"},
    "content": [
        {
            "type": "text/plain",
            "value": "Sales data has been successfully loaded and validated."
        }
    ]
}
}
```

4. Error Handling

```
{
  "name": "Send_Failure_Notification",
  "type": "WebActivity",
  "dependsOn": [
    {
      "activity": "Copy_Sales_Data",
      "dependencyConditions": ["Failed"]
    }
  ],
  "typeProperties": {
    "method": "POST",
    "url": "https://hooks.slack.com/services/...",
    "body": {
      "text": "ALERT: Sales data load failed. Please check ADF monitoring."
    }
  }
}
```

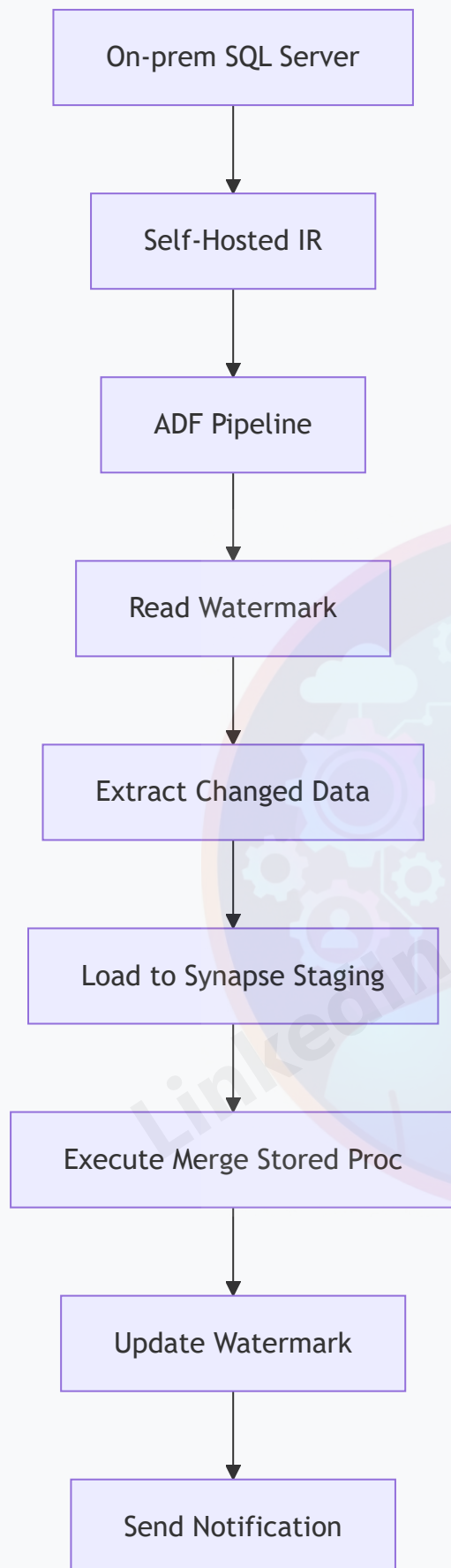
Project 2: Incremental Data Load with Watermarking

Business Scenario

A healthcare provider needs to incrementally load patient records from an on-premises SQL Server to Azure Synapse Analytics, processing only new or changed records since the last load.

Solution Architecture





Implementation Steps

1. Watermark Table Creation

```
CREATE TABLE dbo.WatermarkTable (  
    TableName VARCHAR(100) PRIMARY KEY,  
    LastLoadDate DATETIME2  
);  
  
INSERT INTO dbo.WatermarkTable VALUES  
('Patients', '2024-01-01 00:00:00');
```

2. Pipeline Parameters

```
{  
  "parameters": {  
    "SourceTable": {"type": "string", "defaultValue": "Patients"},  
    "WatermarkColumn": {"type": "string", "defaultValue": "LastModifiedDate"},  
    "BatchSize": {"type": "int", "defaultValue": 10000}  
  }  
}
```

3. Lookup Watermark Activity

```
{  
  "name": "Lookup_Last_Watermark",  
  "type": "Lookup",  
  "typeProperties": {  
    "source": {  
      "type": "SqlServerSource",  
      "sqlReaderQuery": "SELECT LastLoadDate FROM dbo.WatermarkTable WHERE TableName =  
'@{pipeline().parameters.SourceTable}'"  
    },  
    "dataset": {"referenceName": "DS_Watermark_SQL"}  
  }  
}
```

4. Copy Changed Data Activity

```
{  
  "name": "Copy_Incremental_Data",  
  "type": "Copy",  
  "typeProperties": {  
    "source": {  
      "type": "SqlServerSource",  

```

```

    "sqlReaderQuery": "@concat('SELECT * FROM ', pipeline().parameters.SourceTable, '
WHERE ', pipeline().parameters.WatermarkColumn, ' > ''',
activity('Lookup_Last_Watermark').output.firstRow.LastLoadDate, '' ORDER BY ',
pipeline().parameters.WatermarkColumn)"
  },
  "sink": {
    "type": "SqlDWSink",
    "writeBehavior": "insert",
    "preCopyScript": "TRUNCATE TABLE Staging.Patients"
  }
}
}

```

5. Update Watermark Activity

```

{
  "name": "Update_Watermark",
  "type": "SqlServerStoredProcedure",
  "typeProperties": {
    "storedProcedureName": "usp_UpdateWatermark",
    "storedProcedureParameters": {
      "TableName": {"value": "@pipeline().parameters.SourceTable", "type": "String"},
      "NewWatermark": {"value": "@utcnow()", "type": "DateTime"}
    }
  }
}

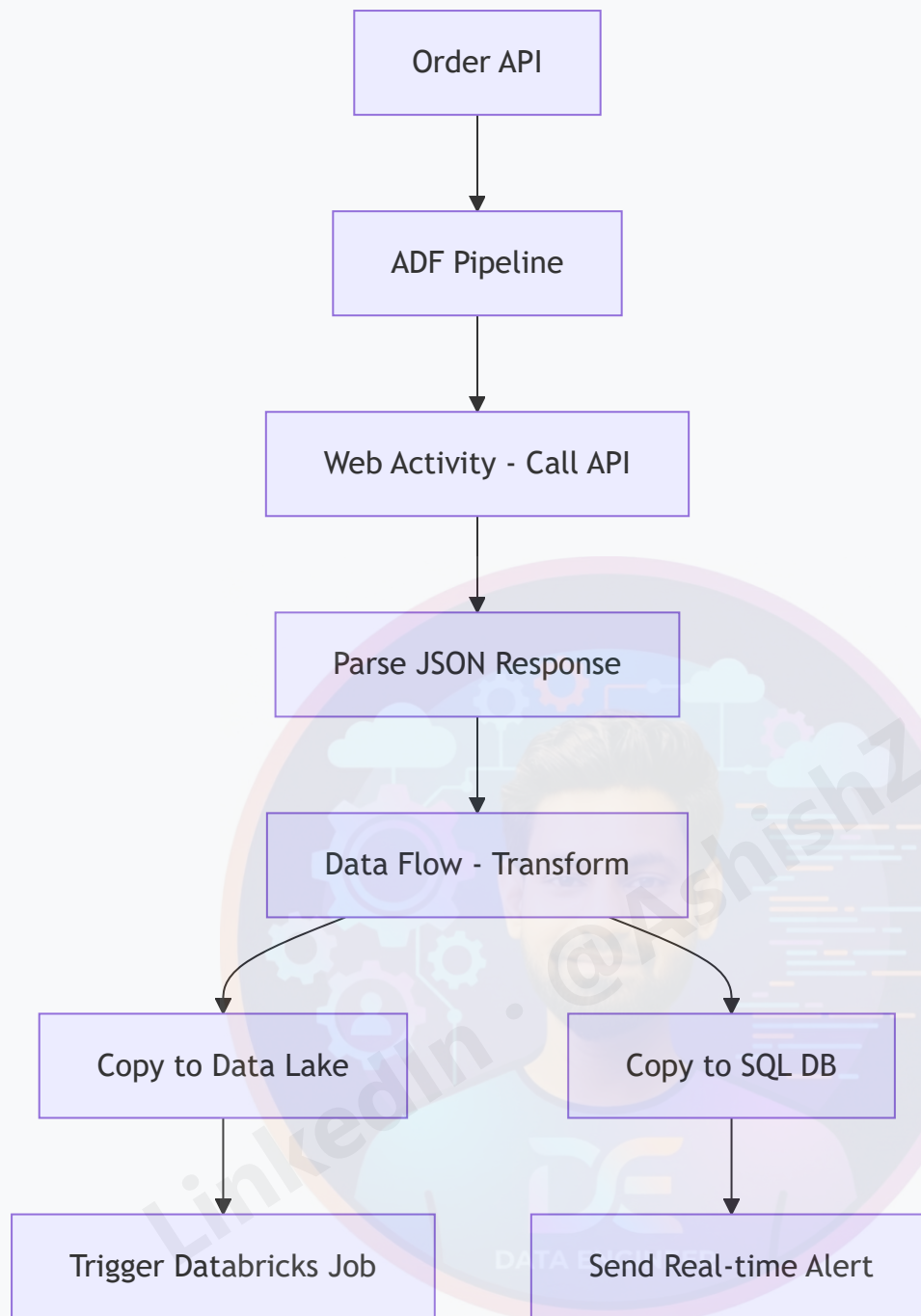
```

Project 3: Real-Time API Data Ingestion

Business Scenario

An e-commerce platform needs to ingest real-time order data from a REST API, transform it, and load it into both a data lake for analytics and a SQL database for operational reporting.

Architecture Overview



Implementation Details

1. REST API Linked Service

```
{
  "name": "LS_Order_API",
  "type": "RestService",
  "typeProperties": {
    "url": "https://api.ecommerce.com/orders",
    "authenticationType": "Basic",
```

```

    "userName": "@linkedService().ApiUsername",
    "password": "@linkedService().ApiPassword"
  }
}

```

2. Web Activity for API Call

```

{
  "name": "Call_Order_API",
  "type": "WebActivity",
  "typeProperties": {
    "method": "GET",
    "url": "https://api.ecommerce.com/orders",
    "headers": {
      "Authorization": "@concat('Bearer ', linkedService().ApiToken)",
      "Content-Type": "application/json"
    },
    "body": {
      "startDate": "@formatDateTime(addhours(utcnow(), -1), 'yyyy-MM-ddTHH:mm:ssZ')",
      "endDate": "@formatDateTime(utcnow(), 'yyyy-MM-ddTHH:mm:ssZ')"
    }
  }
}

```

3. Data Flow for Transformation

```

{
  "name": "DF_Transform_Order_Data",
  "type": "MappingDataFlow",
  "typeProperties": {
    "sources": [
      {
        "name": "OrderSource",
        "dataset": {"referenceName": "DS_Order_JSON"}
      }
    ],
    "transformations": [
      {
        "name": "Flatten_Order_Items",
        "type": "Flatten",
        "typeProperties": {
          "unrollBy": "orderItems",
          "unrollRoot": "orders"
        }
      },
      {
        "name": "Add_Calculated_Fields",
        "type": "DerivedColumn",

```

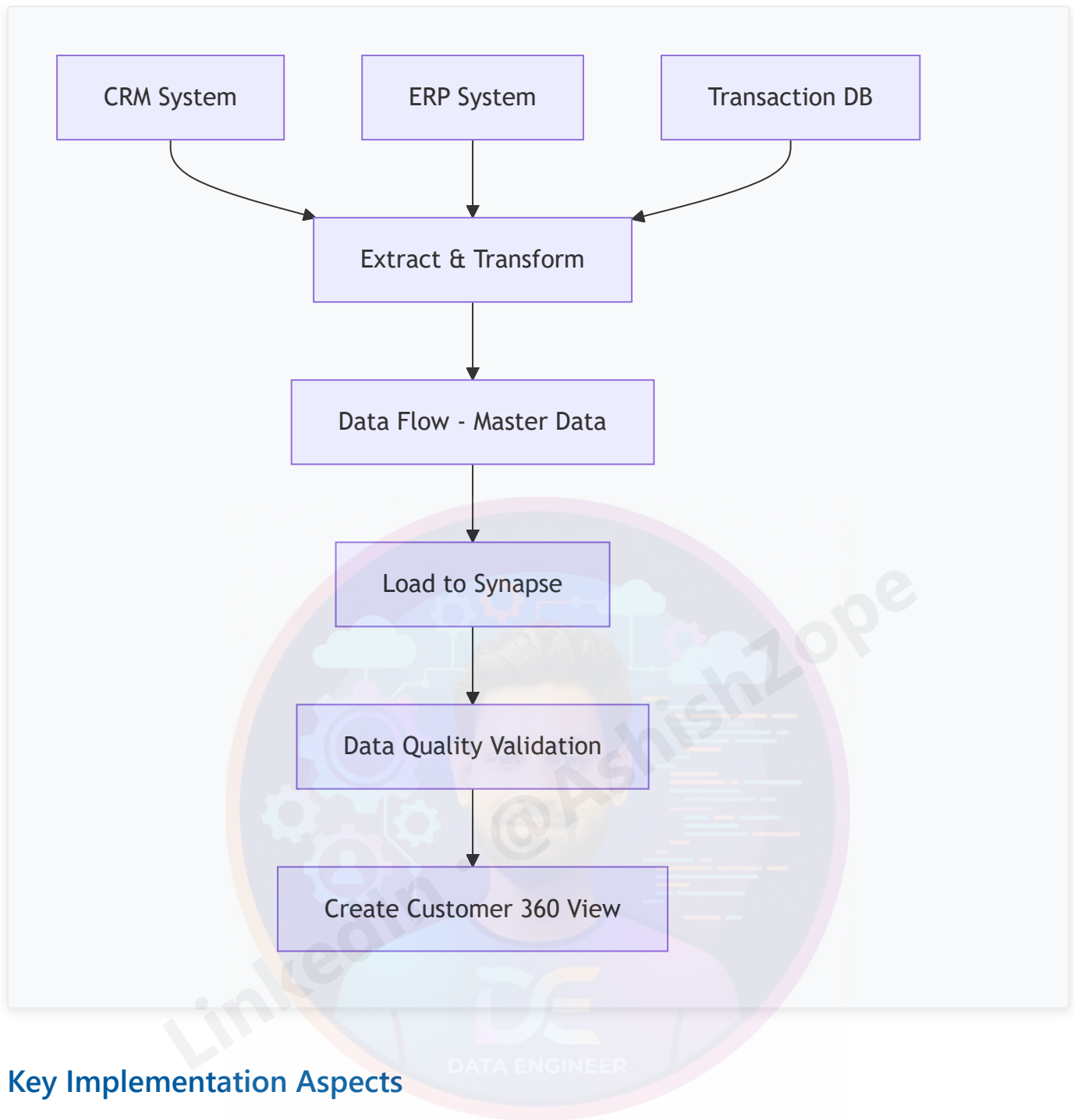
```
"typeProperties": {
  "columns": [
    {
      "name": "OrderTotal",
      "expression": "orderItems.price * orderItems.quantity"
    },
    {
      "name": "ProcessingTimestamp",
      "expression": "currentTimestamp()"
    }
  ]
}
],
"sinks": [
  {
    "name": "DataLakeSink",
    "dataset": {"referenceName": "DS_Order_Parquet"}
  },
  {
    "name": "SqlSink",
    "dataset": {"referenceName": "DS_Order_SQL"}
  }
]
}
```

Project 4: Multi-Source Data Consolidation

Business Scenario

A financial institution needs to consolidate customer data from multiple sources (CRM, ERP, Transaction Systems) into a unified customer view in Azure Synapse.

Solution Approach



Key Implementation Aspects

1. Multiple Source Handling

- Different data formats (JSON, XML, CSV)
- Various connection types (API, Database, Files)
- Incremental vs full load strategies

2. Data Quality Framework

- Duplicate detection and merging
- Data validation rules
- Audit trail maintenance

3. Master Data Management

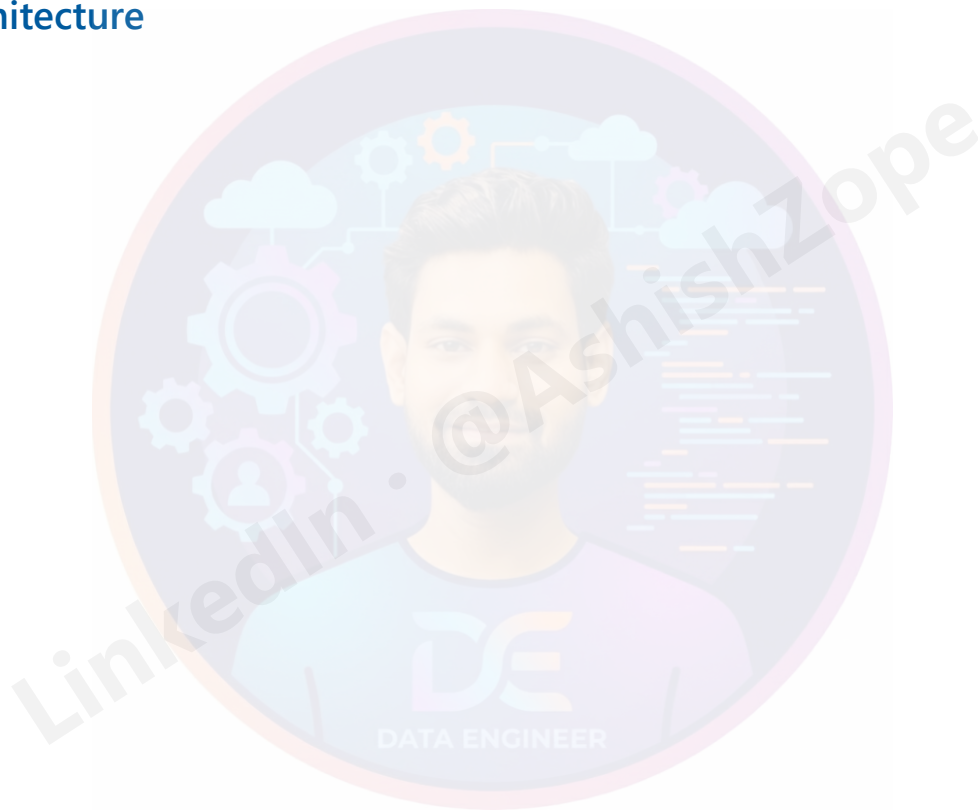
- Customer matching algorithms
- Golden record creation
- Change data capture

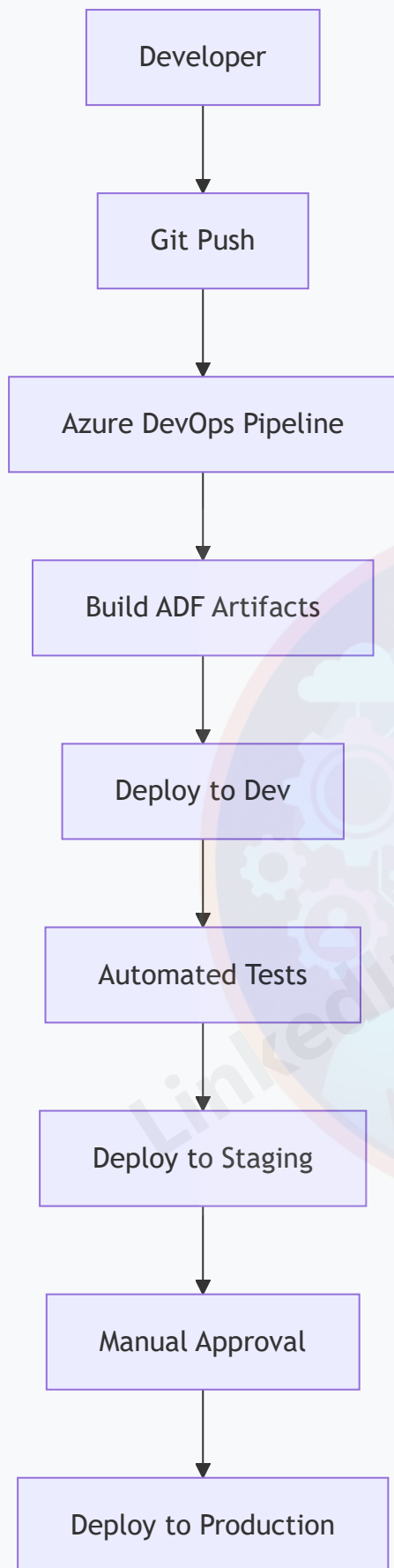
Project 5: CI/CD Pipeline for ADF

Business Scenario

Implement automated deployment of ADF pipelines across development, staging, and production environments with proper testing and approval workflows.

CI/CD Architecture





Implementation Components

1. ARM Template Generation

```
# PowerShell script to export ARM templates
$resourceGroupName = "adf-dev-rg"
$dataFactoryName = "adf-dev-factory"
$armTemplateFolder = "./arm-templates"

Export-AzResourceGroup `
  -ResourceGroupName $resourceGroupName `
  -Resource
"/subscriptions/.../resourceGroups/$resourceGroupName/providers/Microsoft.DataFactory/factories/$dataFactoryName" `
  -Path $armTemplateFolder `
  -Force
```

2. Azure DevOps Pipeline

```
# azure-pipelines.yml
trigger:
  branches:
    include:
      - main
      - develop

stages:
- stage: Build
  jobs:
  - job: BuildADF
    steps:
    - task: AzureResourceManagerTemplateDeployment@3
      inputs:
        deploymentScope: 'Resource Group'
        azureResourceManagerConnection: 'Azure-DevOps-Service-Connection'
        subscriptionId: '$(subscriptionId)'
        action: 'Create Or Update Resource Group'
        resourceGroupName: '$(resourceGroupName)'
        location: '$(location)'
        templateLocation: 'Linked artifact'
        csmFile: '$(System.DefaultWorkingDirectory)/arm-templates/template.json'
        csmParametersFile: '$(System.DefaultWorkingDirectory)/arm-templates/parameters.json'
        overrideParameters: >
          -factoryName "$(factoryName)"
          -environment "$(environment)"

- stage: Test
  condition: and(succeeded(), eq(variables['Build.SourceBranch'], 'refs/heads/develop'))
  jobs:
  - job: IntegrationTests
```

```

steps:
- task: AzureCLI@2
  inputs:
    azureSubscription: 'Azure-DevOps-Service-Connection'
    scriptType: 'bash'
    scriptLocation: 'inlineScript'
    inlineScript: |
      # Run integration tests
      echo "Running ADF integration tests..."

- stage: DeployStaging
  condition: and(succeeded(), eq(variables['Build.SourceBranch'], 'refs/heads/main'))
  jobs:
  - deployment: DeployStaging
    environment: 'staging'
    strategy:
      runOnce:
        deploy:
          steps:
          - task: AzureResourceManagerTemplateDeployment@3
            inputs:
              deploymentScope: 'Resource Group'
              azureResourceManagerConnection: 'Azure-Staging-Service-Connection'
              subscriptionId: '$(stagingSubscriptionId)'
              action: 'Create Or Update Resource Group'
              resourceGroupName: '$(stagingResourceGroupName)'
              location: '$(location)'
              templateLocation: 'Linked artifact'
              csmFile: '$(System.DefaultWorkingDirectory)/arm-templates/template.json'
              csmParametersFile: '$(System.DefaultWorkingDirectory)/arm-
templates/parameters-staging.json'

- stage: DeployProduction
  condition: and(succeeded(), eq(variables['Build.SourceBranch'], 'refs/heads/main'))
  jobs:
  - deployment: DeployProduction
    environment: 'production'
    strategy:
      runOnce:
        deploy:
          steps:
          - task: AzureResourceManagerTemplateDeployment@3
            inputs:
              deploymentScope: 'Resource Group'
              azureResourceManagerConnection: 'Azure-Prod-Service-Connection'
              subscriptionId: '$(prodSubscriptionId)'
              action: 'Create Or Update Resource Group'
              resourceGroupName: '$(prodResourceGroupName)'
              location: '$(location)'
              templateLocation: 'Linked artifact'
              csmFile: '$(System.DefaultWorkingDirectory)/arm-templates/template.json'
              csmParametersFile: '$(System.DefaultWorkingDirectory)/arm-
templates/parameters-prod.json'

```

3. Parameter Files for Different Environments

```
// parameters-dev.json
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "factoryName": {
      "value": "adf-dev-datafactory"
    },
    "LS_SQL_connectionString": {
      "value": "Server=tcp:dev-sql-server.database.windows.net,1433;Database=dev-db;"
    },
    "LS_Blob_connectionString": {
      "value": "DefaultEndpointsProtocol=https;AccountName=devstorage;AccountKey=...;"
    }
  }
}
```

Best Practices for End-to-End Projects

Project Planning

1. **Requirements Gathering:** Understand business needs and technical constraints
2. **Architecture Design:** Design scalable and maintainable solutions
3. **Data Modeling:** Plan data structures and relationships
4. **Security Planning:** Implement appropriate security measures

Development Best Practices

1. **Modular Design:** Break complex pipelines into reusable components
2. **Error Handling:** Implement comprehensive error handling and logging
3. **Testing Strategy:** Unit tests, integration tests, and performance tests
4. **Documentation:** Maintain detailed documentation for maintenance

Deployment Best Practices

1. **Environment Strategy:** Dev → Staging → Production progression
2. **CI/CD Pipeline:** Automated testing and deployment
3. **Configuration Management:** Environment-specific configurations
4. **Rollback Strategy:** Ability to quickly rollback changes

Monitoring and Maintenance

1. **Monitoring Setup:** Comprehensive monitoring and alerting
 2. **Performance Tuning:** Regular performance optimization
 3. **Security Updates:** Keep components up to date
 4. **Documentation Updates:** Maintain current documentation
-

46. Best Practices

Development Best Practices

- Use naming conventions
- Use parameterization
- Avoid hardcoding
- Implement logging
- Use reusable pipelines
- Use Git integration
- Enable monitoring

Performance Best Practices

- Use partitioning
 - Use incremental load
 - Optimize data flows
 - Use staging carefully
-

47. Common Interview Questions

Beginner Level

1. What is Azure Data Factory?
2. What is Integration Runtime?
3. Difference between dataset and linked service?
4. What is a pipeline?
5. What are triggers?

Intermediate Level

1. Explain incremental loading.
2. Difference between ETL and ELT?
3. Explain mapping data flow.
4. How does self-hosted IR work?
5. Explain parameterization.

Advanced Level

1. How to optimize ADF performance?
2. Explain CI/CD in ADF.
3. How to handle pipeline failures?
4. Explain dynamic content.
5. How to secure ADF pipelines?

48. ADF Developer Roadmap

Phase 1: Basics

- Azure basics
- Cloud concepts
- Storage concepts
- SQL fundamentals

Phase 2: ADF Fundamentals

- Pipelines
- Activities
- Datasets
- Linked services
- Triggers

Phase 3: Intermediate

- Data flow
- Dynamic content
- Incremental load

- API integration
- Error handling

Phase 4: Advanced

- CI/CD
- Databricks integration
- Synapse integration
- Security
- Performance tuning

Phase 5: Real Projects

- End-to-end projects
 - Production scenarios
 - Monitoring implementation
-

49. ADF Naming Standards

Recommended Naming Conventions

COMPONENT	NAMING EXAMPLE
Pipeline	PL_Copy_Customer
Dataset	DS_SQL_Employee
Linked Service	LS_SQL_Server
Trigger	TR_Daily_Load
Data Flow	DF_Transform_Sales

50. Important Terminologies

TERM	MEANING
ETL	Extract Transform Load
ELT	Extract Load Transform

TERM	MEANING
IR	Integration Runtime
CDC	Change Data Capture
DIU	Data Integration Unit
RBAC	Role-Based Access Control
CI/CD	Continuous Integration/Deployment

53. Troubleshooting and Common Issues

Pipeline Execution Issues

Issue: Pipeline Runs But Activities Fail

Symptoms:

- Pipeline status shows "Succeeded" but activities fail
- Inconsistent execution results
- Activities show "Skipped" status

Possible Causes:

1. **Dependency Conditions:** Incorrect dependency configuration
2. **Error Handling:** Activities set to continue on failure
3. **Trigger Configuration:** Trigger not properly configured

Solutions:

```
// Check dependency conditions
"dependsOn": [
  {
    "activity": "CopyData",
    "dependencyConditions": ["Succeeded"] // Change to ["Completed"] if needed
  }
]

// Enable proper error propagation
"policy": {
  "timeout": "01:00:00",
  "retry": 3,
```



```
"retryIntervalInSeconds": 30
}
```

Issue: Copy Activity Performance Degradation

Symptoms:

- Copy jobs taking longer than expected
- High DIU usage but slow throughput
- Pipeline timeouts

Troubleshooting Steps:

1. **Check DIU Usage:** Monitor Data Integration Unit consumption
2. **Review Source/Target:** Check if source or target is bottleneck
3. **Examine Data Skew:** Look for uneven data distribution
4. **Verify Network:** Check network bandwidth and latency

Optimization Solutions:

```
// Increase parallel copies
"typeProperties": {
  "parallelCopies": 8,
  "dataIntegrationUnits": 32
}

// Use staging for cross-region transfers
"enableStaging": true,
"stagingSettings": {
  "linkedServiceName": {
    "referenceName": "LS_StagingStorage"
  }
}
```

Connection and Authentication Issues

Issue: Linked Service Connection Failures

Common Error Messages:

- "Login failed for user"
- "Cannot open server connection"
- "Access denied"

Troubleshooting Checklist:

1. **Credentials Validation:** Verify username/password or keys
2. **Firewall Rules:** Check if ADF IP ranges are whitelisted
3. **Network Connectivity:** Test connection from ADF region
4. **Permissions:** Ensure proper database/object permissions

Resolution Examples:

```
// For Azure SQL Database
{
  "type": "AzureSqlDatabase",
  "typeProperties": {
    "connectionString":
"Server=tcp:server.database.windows.net,1433;Database=db;Authentication=Active Directory
Password;Encrypt=True;",
    "password": {
      "type": "AzureKeyVaultSecret",
      "store": {
        "referenceName": "LS_KeyVault"
      },
      "secretName": "sql-password"
    }
  }
}
```

Issue: Self-Hosted IR Connectivity Problems

Symptoms:

- IR shows "Offline" status
- Pipeline fails with connectivity errors
- Intermittent connection issues

Diagnostic Steps:

1. **IR Service Status:** Check if SHIR service is running
2. **Network Configuration:** Verify firewall and proxy settings
3. **DNS Resolution:** Test DNS resolution from SHIR machine
4. **Port Availability:** Ensure required ports are open

Configuration Fixes:

```
// SHIR configuration
{
  "name": "SHIR_OnPrem",
  "type": "SelfHosted",
  "typeProperties": {
    "linkedInfos": [
      {

```

```
"name": "OnPremSqlServer",
"type": "SqlServer",
"typeProperties": {
  "connectionString": "Server=sqlserver.company.com;Database=ProdDB;",
  "userName": "adf_user",
  "password": {
    "type": "SecureString",
    "value": "****"
  }
}
}
```

Data Flow Issues

Issue: Data Flow Cluster Startup Failures

Symptoms:

- Data flow fails with cluster creation errors
- Long startup times
- Out of memory errors

Solutions:

```
// Optimize cluster configuration
{
  "typeProperties": {
    "compute": {
      "coreCount": 8,
      "computeType": "General"
    },
    "timeToLive": 10
  }
}
```

Issue: Data Skew and Performance Issues

Symptoms:

- Uneven data distribution across partitions
- Some partitions processing much slower
- Memory pressure on certain nodes

Optimization Techniques:

```
// Implement proper partitioning
{
  "partitioning": {
    "type": "Hash",
    "partitionColumnName": "customer_id"
  }
}

// Use broadcast join for small tables
{
  "join": {
    "type": "Broadcast"
  }
}
```

Trigger and Scheduling Issues

Issue: Triggers Not Firing

Symptoms:

- Scheduled pipelines not executing
- Event-based triggers not responding
- Tumbling window triggers missing data

Diagnostic Queries:

```
-- Check trigger runs in ADF monitoring
SELECT * FROM dbo.TriggerRuns
WHERE TriggerName = 'MyTrigger'
ORDER BY RunStart DESC

-- Check for trigger failures
SELECT * FROM dbo.ActivityRuns
WHERE PipelineName = 'MyPipeline'
AND Status = 'Failed'
```

Common Fixes:

```
// Fix schedule trigger timezone issues
{
  "type": "ScheduleTrigger",
```

```
"typeProperties": {  
  "recurrence": {  
    "frequency": "Day",  
    "interval": 1,  
    "startTime": "2024-01-01T06:00:00Z",  
    "timeZone": "UTC"  
  }  
}
```

Memory and Resource Issues

Issue: Out of Memory Errors

Symptoms:

- Pipeline fails with OOM errors
- Data flow cluster crashes
- Activities timeout unexpectedly

Memory Optimization:

```
// Increase cluster size for data flows  
{  
  "compute": {  
    "coreCount": 16,  
    "computeType": "MemoryOptimized"  
  }  
}  
  
// Implement paging for large datasets  
{  
  "source": {  
    "type": "SqlServerSource",  
    "isolationLevel": "ReadUncommitted",  
    "partitionOption": "DynamicRange",  
    "partitionSettings": {  
      "partitionColumnName": "id",  
      "partitionUpperBound": "1000000",  
      "partitionLowerBound": "1"  
    }  
  }  
}
```

Debugging Techniques

Using Debug Mode

```
// Enable debug mode for pipelines
{
  "debug": true,
  "parameters": {
    "debugDataset": {
      "value": "DS_Debug_Output"
    }
  }
}
```

Logging Best Practices

```
// Implement comprehensive logging
{
  "activities": [
    {
      "name": "Log_Start",
      "type": "SetVariable",
      "typeProperties": {
        "variableName": "ExecutionLog",
        "value": "@concat('Pipeline started at: ', utcnow())"
      }
    },
    {
      "name": "Log_Activity_Result",
      "type": "AppendVariable",
      "typeProperties": {
        "variableName": "ExecutionLog",
        "value": "@concat(activity('CopyData').output.rowsCopied, ' rows copied')"
      }
    }
  ]
}
```

Monitoring Queries

```
-- Pipeline performance monitoring
SELECT
  PipelineName,
```

```
AVG(DurationInMs) as AvgDuration,
MAX(DurationInMs) as MaxDuration,
COUNT(*) as TotalRuns,
SUM(CASE WHEN Status = 'Failed' THEN 1 ELSE 0 END) as FailedRuns
FROM dbo.PipelineRuns
WHERE RunStart > DATEADD(day, -30, GETDATE())
GROUP BY PipelineName
ORDER BY AvgDuration DESC

-- Activity error analysis
SELECT
    ActivityName,
    ErrorMessage,
    COUNT(*) as ErrorCount,
    MAX(RunStart) as LastError
FROM dbo.ActivityRuns
WHERE Status = 'Failed'
AND RunStart > DATEADD(day, -7, GETDATE())
GROUP BY ActivityName, ErrorMessage
ORDER BY ErrorCount DESC
```

Performance Tuning Checklist

Data Movement Optimization

- ☐ Use appropriate DIU settings
- ☐ Enable parallel copying
- ☐ Implement data partitioning
- ☐ Use staging for large transfers
- ☐ Compress data when possible

Data Flow Optimization

- ☐ Choose correct compute type
- ☐ Implement proper partitioning
- ☐ Use caching strategically
- ☐ Optimize join types
- ☐ Monitor cluster utilization

Pipeline Optimization

- ☐ Minimize activity count
- ☐ Use parallel execution
- ☐ Implement incremental loads
- ☐ Optimize trigger schedules

- ☐ Monitor resource usage

Infrastructure Optimization

- ☐ Choose correct IR type
- ☐ Optimize network configuration
- ☐ Implement proper security
- ☐ Monitor costs and usage
- ☐ Plan for scaling needs

52. Certification Guidance

Recommended Certifications

Microsoft Certifications

- DP-203: Data Engineering on Microsoft Azure
- AZ-900: Azure Fundamentals
- DP-900: Data Fundamentals

Preparation Strategy

1. Learn basics
2. Build projects
3. Practice labs
4. Solve interview questions
5. Attempt mock tests

54. Advanced Topics and Future Trends

Advanced ADF Features

1. Change Data Capture (CDC) Patterns

Native CDC with Azure SQL DB

```

{
  "name": "PL_CDC_Load",
  "properties": {
    "activities": [
      {
        "name": "Get_Change_Data",
        "type": "Lookup",
        "typeProperties": {
          "source": {
            "type": "AzureSqlSource",
            "sqlReaderQuery": "
SELECT * FROM cdc.dbo_Customers_CT
WHERE __$start_lsn > @last_lsn
AND __$operation IN (1, 2, 4) -- Insert, Update, Delete
ORDER BY __$start_lsn"
          },
          "dataset": {"referenceName": "DS_SQL_Customers"}
        }
      },
      {
        "name": "Process_Changes",
        "type": "ForEach",
        "typeProperties": {
          "items": "@activity('Get_Change_Data').output.value",
          "activities": [
            {
              "name": "Route_By_Operation",
              "type": "Switch",
              "typeProperties": {
                "on": "@item().__operation",
                "cases": {
                  "1": [{"name": "Insert_Record", "type": "Copy"}],
                  "2": [{"name": "Update_Record", "type": "Copy"}],
                  "4": [{"name": "Delete_Record", "type": "SqlServerStoredProcedure"}]
                }
              }
            }
          ]
        }
      }
    ]
  }
}

```

Custom CDC Implementation

```

-- Watermark table for custom CDC
CREATE TABLE dbo.CDC_Watermark (
  TableName VARCHAR(100) PRIMARY KEY,

```



```

        LastLSN BINARY(10),
        LastProcessedDate DATETIME2
    );

-- Incremental load with custom logic
SELECT *
FROM Customers
WHERE LastModifiedDate > (
    SELECT LastProcessedDate
    FROM dbo.CDC_Watermark
    WHERE TableName = 'Customers'
)
AND LastModifiedDate <= GETDATE();

```

2. Event-Driven Architectures

Event Grid Integration

```

{
  "name": "TR_Event_Trigger",
  "properties": {
    "type": "BlobEventsTrigger",
    "typeProperties": {
      "blobPathBeginsWith": "/data/input/",
      "blobPathEndsWith": ".csv",
      "ignoreEmptyBlobs": true,
      "scope":
"/subscriptions/.../resourceGroups/.../providers/Microsoft.Storage/storageAccounts/...",
      "events": ["Microsoft.Storage.BlobCreated"]
    },
    "pipelines": [
      {
        "pipelineReference": {
          "referenceName": "PL_Process_File",
          "type": "PipelineReference"
        },
        "parameters": {
          "fileName": "@triggerBody().fileName",
          "folderPath": "@triggerBody().folderPath"
        }
      }
    ]
  }
}

```

Real-Time Data Processing

```

{
  "name": "PL_RealTime_Processing",
  "properties": {
    "activities": [
      {
        "name": "Ingest_Stream_Data",
        "type": "Copy",
        "typeProperties": {
          "source": {
            "type": "EventHubSource",
            "consumerGroupName": "$Default"
          },
          "sink": {
            "type": "SqlSink",
            "writeBehavior": "upsert"
          }
        }
      },
      {
        "name": "Trigger_Stream_Analytics",
        "type": "WebActivity",
        "typeProperties": {
          "method": "POST",
          "url": "https://management.azure.com/.../start?api-version=2016-03-01",
          "authentication": {
            "type": "MSI",
            "resource": "https://management.azure.com/"
          }
        }
      }
    ]
  }
}

```

3. Machine Learning Integration

ML Model Scoring in Data Flows

```

{
  "name": "DF_ML_Scoring",
  "type": "MappingDataFlow",
  "typeProperties": {
    "sources": [
      {
        "name": "InputData",
        "dataset": {"referenceName": "DS_Input_Data"}
      }
    ],
  }
}

```

```

"transformations": [
  {
    "name": "Call_ML_Model",
    "type": "DerivedColumn",
    "typeProperties": {
      "columns": [
        {
          "name": "Prediction",
          "expression": "amlPredict('workspace/model', input_data)"
        }
      ]
    }
  }
],
"sinks": [
  {
    "name": "ScoredData",
    "dataset": {"referenceName": "DS_Scored_Output"}
  }
]
}

```

Automated ML Pipeline

```

{
  "name": "PL_AutoML_Training",
  "properties": {
    "activities": [
      {
        "name": "Prepare_Training_Data",
        "type": "DataFlow",
        "typeProperties": {
          "dataFlow": {"referenceName": "DF_Data_Preparation"}
        }
      },
      {
        "name": "Train_ML_Model",
        "type": "AzureMLExecutePipeline",
        "typeProperties": {
          "experimentName": "Customer_Churn_Prediction",
          "mlPipelineId": "pipeline_id",
          "mlPipelineParameters": {
            "training_data": "@activity('Prepare_Training_Data').output"
          }
        }
      },
      {
        "name": "Register_Model",
        "type": "AzureMLExecutePipeline",
        "dependsOn": [

```

```
{
  {
    "activity": "Train_ML_Model",
    "dependencyConditions": ["Succeeded"]
  }
}
]
```

Future Trends in ADF

1. AI-Powered Data Integration

Intelligent Mapping

- **Auto-mapping:** AI suggests column mappings based on data patterns
- **Schema Evolution:** Automatic handling of schema changes
- **Data Quality Scoring:** ML-based data quality assessment

Predictive Optimization

- **Auto-tuning:** AI recommends performance optimizations
- **Cost Optimization:** Intelligent resource allocation
- **Anomaly Detection:** ML-based pipeline failure prediction

2. Low-Code/No-Code Evolution

Visual Pipeline Builder

- **Drag-and-drop Enhancements:** More intuitive interface
- **Template Library:** Pre-built pipeline templates
- **Copilot Integration:** AI-assisted pipeline development

Citizen Integrator Features

- **Natural Language Processing:** Describe pipelines in plain English
- **Auto-generated Documentation:** Automatic pipeline documentation
- **Business User Tools:** Simplified interface for business users

3. Multi-Cloud and Hybrid Evolution

Cross-Cloud Data Movement

```
// Future cross-cloud linked service
{
  "name": "LS_AWS_S3_CrossCloud",
  "type": "AmazonS3",
  "typeProperties": {
    "cloudType": "AWS",
    "region": "us-east-1",
    "crossCloudEnabled": true
  }
}
```

Unified Control Plane

- **Single Pane of Glass:** Manage all cloud resources from one interface
- **Unified Security:** Consistent security policies across clouds
- **Global Optimization:** Intelligent data placement and movement

4. Real-Time and Streaming Integration

Native Streaming Support

- **Event-Driven Pipelines:** React to events in real-time
- **Stream Processing:** Built-in stream processing capabilities
- **Change Streams:** Direct integration with database change streams

Advanced Analytics Integration

- **Real-Time ML Scoring:** Stream data through ML models
- **Complex Event Processing:** Pattern detection and alerting
- **Time-Series Analytics:** Built-in time-series processing

5. Serverless and Event-Driven Architecture

Serverless Execution

- **Pay-per-execution:** Only pay for actual pipeline runs
- **Auto-scaling:** Instant scaling based on workload
- **Event-driven:** Trigger pipelines based on any event

Microservices Integration

- **Container Support:** Run pipelines in containers

- **Kubernetes Integration:** Native K8s support
- **Service Mesh:** Advanced service-to-service communication

6. Advanced Security and Compliance

Zero-Trust Architecture

- **Continuous Verification:** Always-on security validation
- **Fine-grained Access:** Row-level and column-level security
- **Data Masking:** Automatic PII detection and masking

Compliance Automation

- **Audit Automation:** Automated compliance reporting
- **Data Lineage:** End-to-end data tracking
- **Retention Policies:** Automated data lifecycle management

7. Industry-Specific Solutions

Healthcare Data Integration

- **FHIR Integration:** Native Fast Healthcare Interoperability Resources support
- **HIPAA Compliance:** Built-in healthcare compliance features
- **Patient Data Lakes:** Specialized healthcare data architectures

Financial Services

- **Regulatory Reporting:** Automated regulatory compliance
- **Risk Analytics:** Real-time risk assessment pipelines
- **Trade Surveillance:** Automated trade monitoring and alerting

8. Sustainability and Green Computing

Carbon-Aware Computing

- **Green Scheduling:** Schedule jobs during low-carbon periods
- **Energy Optimization:** Optimize resource usage for energy efficiency
- **Carbon Tracking:** Monitor and report carbon footprint

Sustainable Architecture

- **Edge Computing:** Process data closer to source to reduce network traffic
- **Efficient Storage:** Optimize storage patterns for energy efficiency

- **Workload Consolidation:** Intelligent workload placement

Preparing for Future ADF

Skills Development Roadmap

Phase 1: Current Skills (2024-2025)

- Master current ADF features
- Learn advanced data flow patterns
- Understand security and compliance

Phase 2: Emerging Skills (2025-2026)

- AI/ML integration patterns
- Multi-cloud architectures
- Real-time data processing

Phase 3: Future Skills (2026-2027)

- Event-driven architectures
- Serverless data integration
- Sustainable data practices

Technology Adoption Strategy

Pilot Programs

- Start with small, low-risk projects
- Test new features in development environments
- Gather feedback and metrics

Gradual Migration

- Migrate existing pipelines incrementally
- Use feature flags for new capabilities
- Maintain backward compatibility

Continuous Learning

- Follow ADF release notes and blogs
- Participate in preview programs

- Join ADF user communities

Organizational Readiness

Team Skills Assessment

- Evaluate current team capabilities
- Identify skill gaps
- Plan training and certification programs

Process Adaptation

- Update development processes
- Modify testing strategies
- Enhance monitoring and alerting

Governance Framework

- Update data governance policies
- Enhance security frameworks
- Modify compliance procedures

Conclusion

Azure Data Factory continues to evolve rapidly, incorporating the latest trends in data integration, AI/ML, and cloud computing. By staying current with these trends and preparing for future developments, organizations can maximize their investment in ADF and maintain a competitive edge in data integration.

Key takeaways for future readiness:

1. **Embrace AI/ML Integration:** Learn to incorporate machine learning into data pipelines
2. **Adopt Event-Driven Patterns:** Move from batch to real-time processing
3. **Plan for Multi-Cloud:** Design architectures that span multiple cloud providers
4. **Focus on Sustainability:** Consider environmental impact in data architecture decisions
5. **Invest in Skills:** Continuously update team skills and knowledge
6. **Stay Current:** Follow ADF developments and adopt new features strategically

54. Conclusion

Azure Data Factory is one of the most important cloud data integration tools for Data Engineers.

ADF helps organizations:

- Automate workflows

- Move large-scale data
- Build ETL/ELT pipelines
- Integrate multiple systems
- Create scalable cloud architectures

To become strong in ADF:

1. Learn fundamentals properly
2. Build hands-on projects
3. Practice real scenarios
4. Understand cloud concepts
5. Learn SQL deeply
6. Work on incremental loading
7. Understand monitoring and security

ADF combined with:

- SQL
- Python
- PySpark
- Synapse
- Databricks

creates a strong Data Engineering skill set.

Final Learning Sequence

1. Azure Basics
2. SQL Fundamentals
3. Azure Storage
4. Azure Data Factory Basics
5. Pipelines
6. Activities
7. Triggers
8. Dynamic Content
9. Data Flow
10. Incremental Load
11. Error Handling
12. Monitoring
13. Git Integration
14. CI/CD
15. Synapse
16. Databricks
17. End-to-End Projects
18. Interview Preparation
19. Certification

20. Production-Level Architecture

